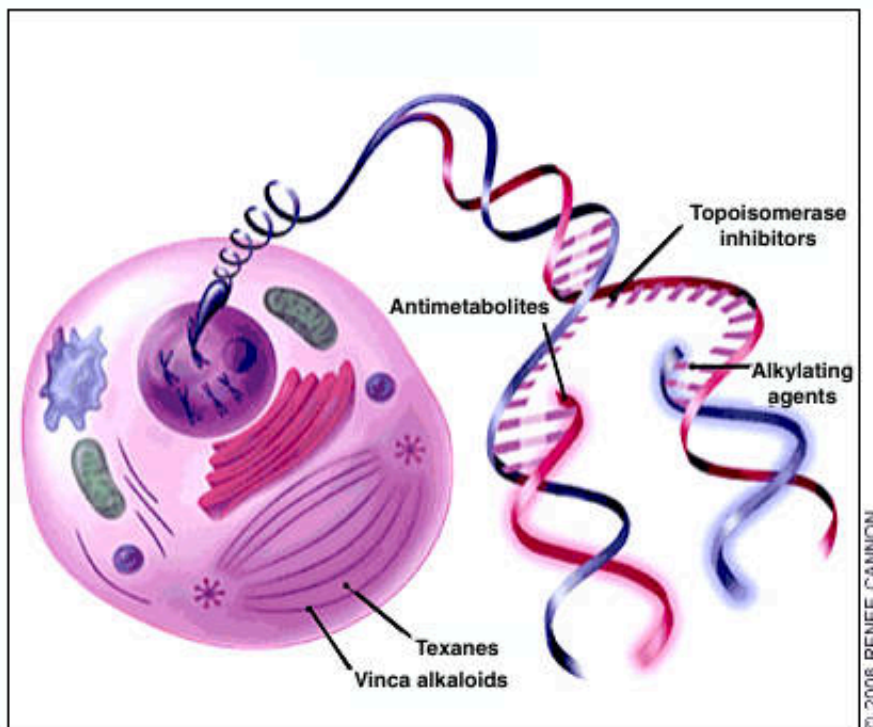




Anticancer agents II: Targeted Therapy

Krisada Sakchaisri, Ph.D.

1



2 Am Fam Physician. 2006 Feb 1;77(3):311-319

Targeted Therapies



Block the growth and spread of cancer by interfering with specific molecules ("**molecular targets**") that are involved in the growth, progression, and spread of cancer

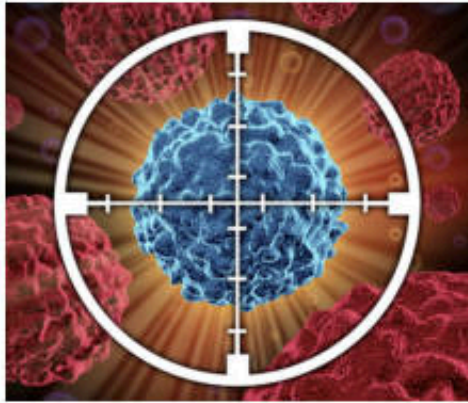
- ❖ Targeting on specific molecules that are associated with cancer
- ❖ Deliberately chosen or designed to interact with the target
- ❖ Often cytostatic

3



4

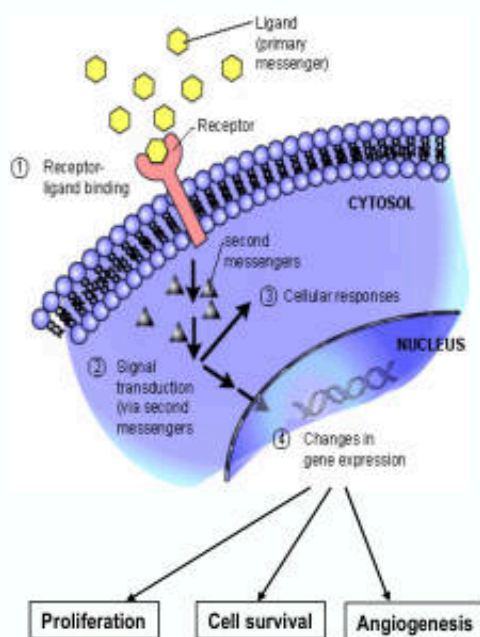
Targeted Therapies



- Signal transduction inhibitors
- Angiogenesis inhibitors
- Immunotherapy
- Apoptosis-inducing agents
- Hormonal therapies
- Gene expression modulators

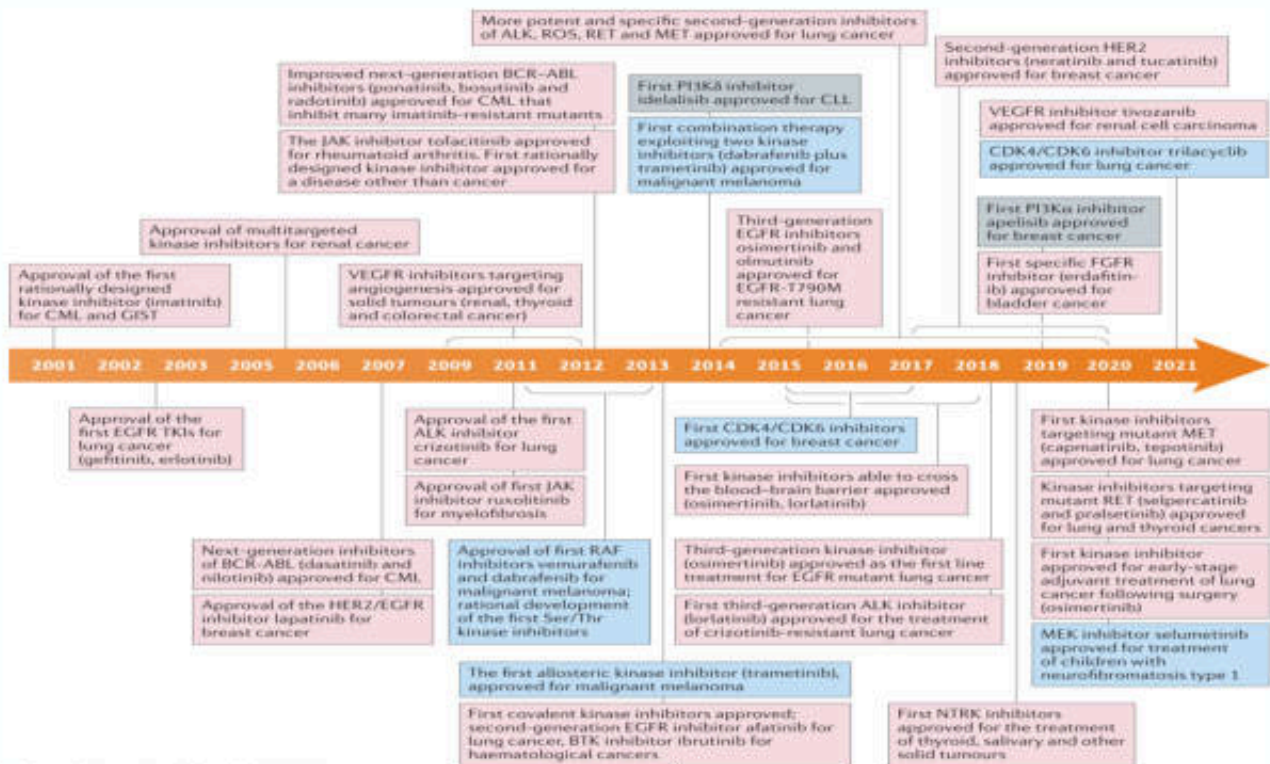
5

Signal Transduction Inhibitors



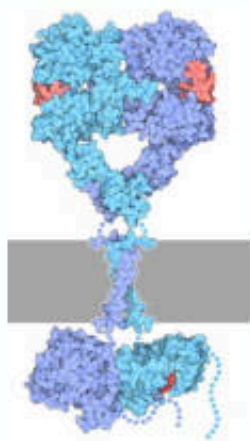
- ✦ Epidermal growth factor receptor (EGFR) inhibitors
- ✦ HER2 inhibitors
- ✦ BCR-ABL inhibitors
- ✦ mTOR inhibitors
- ✦ BRAF inhibitors
- ✦ PI3K inhibitors
- ✦ NTRK inhibitors
- ✦ PARB inhibitors
- ✦ BTK inhibitors
- ✦ JAK inhibitors
- ✦ ALK inhibitors

6



Nat Rev Drug Discov. 2021 Jul;20(7):551-569

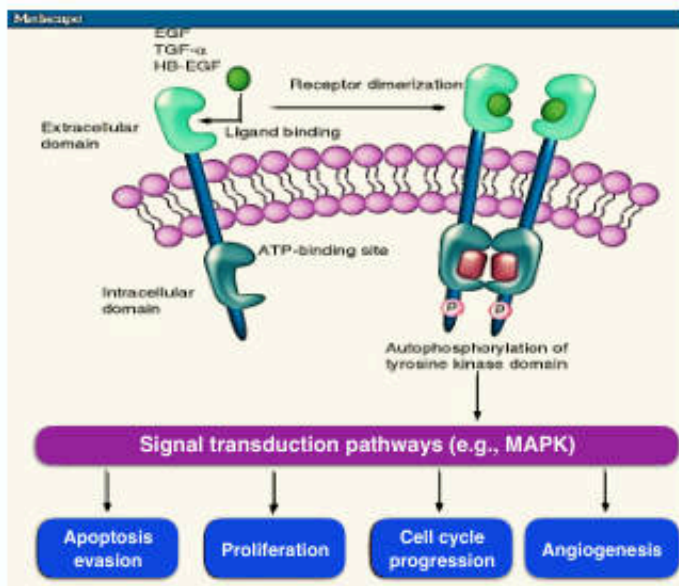
7



EGFR Inhibitors

- ♦ small molecule Inhibitors/tyrosine kinase inhibitors (TKIs)
- ♦ monoclonal antibodies (mAbs)

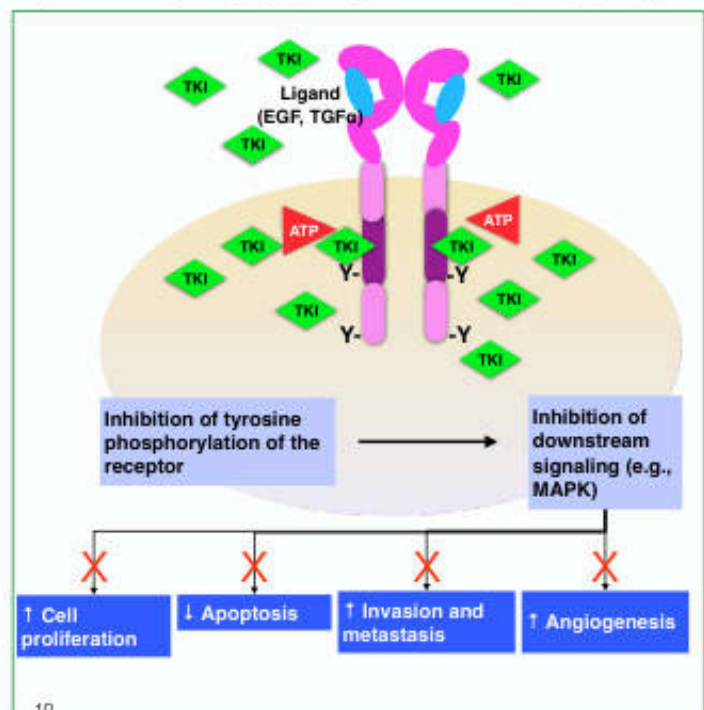
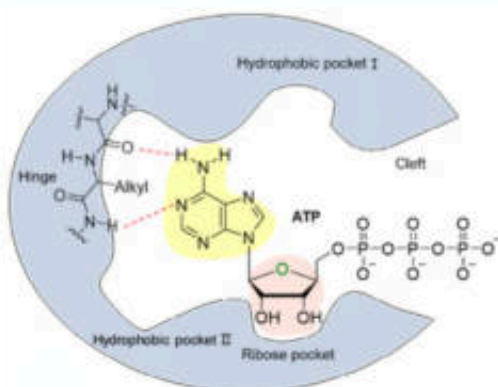
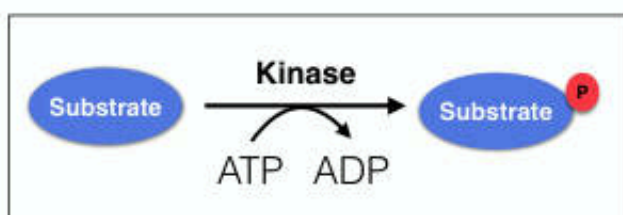
Epidermal Growth Factor Receptor (EGFR)



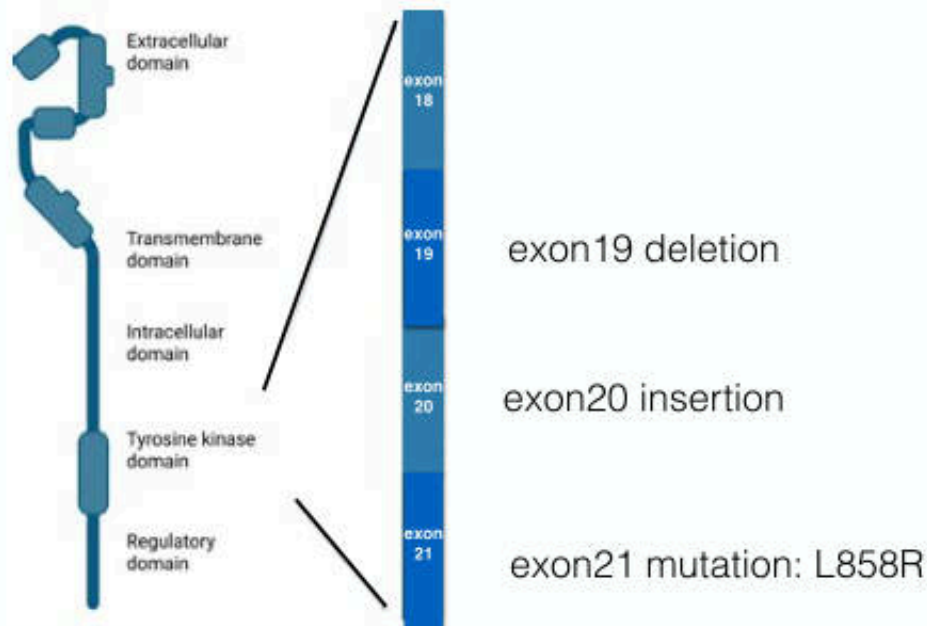
- ✦ also known as ErbB1/HER1, belonging to the ErbB family of transmembrane receptor tyrosine kinases
- ✦ essential for the growth and differentiation of epithelial cells
- ✦ found at abnormally high levels in cancer cells, eg. lung and colorectal cancers

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EGFR Inhibitors: Small Molecule Inhibitors

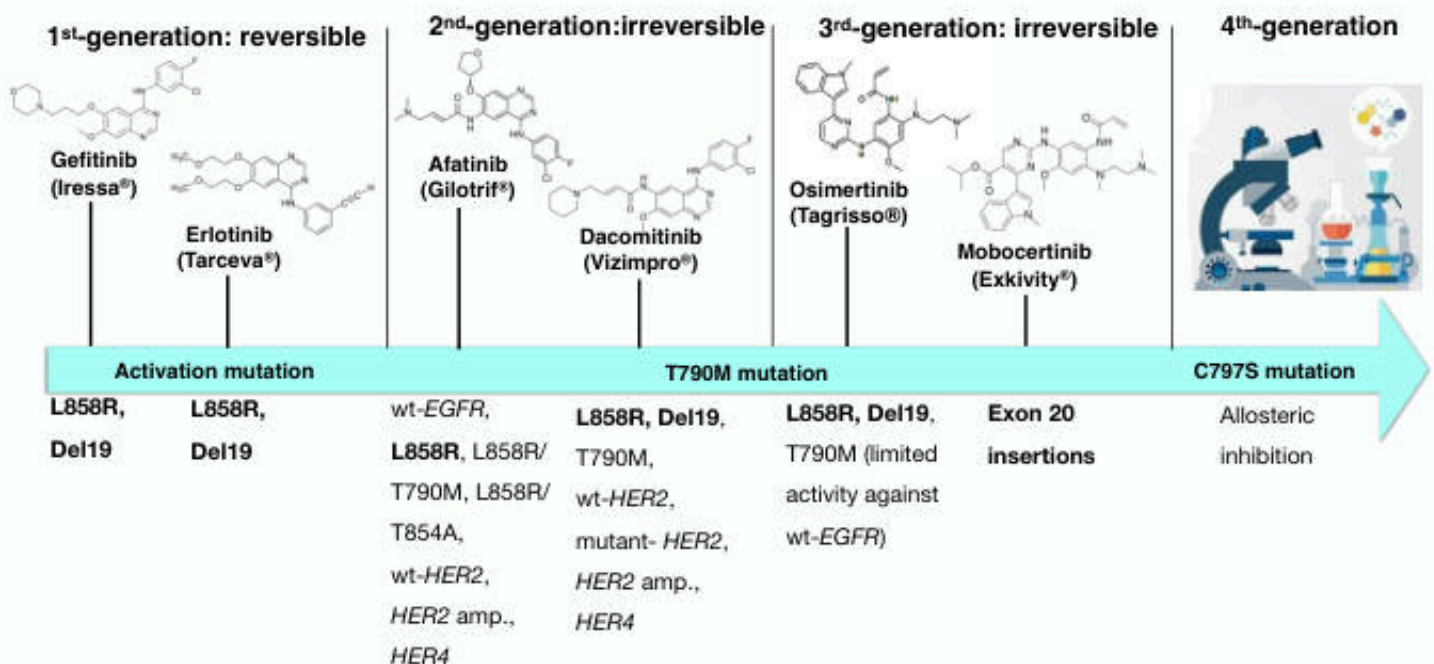


EGFR Mutation in Lung Cancer



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EGFR Inhibitors: Small Molecule Inhibitors



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EGFR Inhibitors in Lung Cancer

Parameter	Erlotinib	Gefitinib	Afatinib	Dacomitinib	Osimertinib
Generation	First	First	Second	Second	Third
EGFR mutations approved for in first-line setting	Ex19del, Ex21 L858R	Ex19del, Ex21 L858R	Ex18 G719X,* Ex19del, Ex20 S768I,* Ex21 L858R, Ex21 L861Q*	Ex19del, Ex21 L858R	Ex19del, Ex21 L858R†
EGFR binding	Reversible	Reversible	Irreversible	Irreversible	Irreversible
CNS Penetration, AUC ratio	0.03X CSF/Plasma	0.01X CSF/Serum	0.02X CSF/Plasma	CNS activity reported	2X brain/plasma

*Uncommon non-resistant EGFR mutations

†Approved for second-line therapy of EGFR exon 20 resistant mutation T790M

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EGFR Inhibitors: Small Molecule Inhibitors

	Gefitinib	Erlotinib	Afatinib	Dacomitinib	Osimertinib	Mobocertinib
F (%)	60	~60	92	80	~70	~37
T _{max} (h)	3-7	4	2-5	6	6	4
Vd/F (L)	1,400	232	2,870	1889	986	3,509
Protein binding (%)	90	~93	95	~98	~95	~99
t _{1/2} (h)	~48	~36	~37	~70	~48	~18
Metabolism	CYP3A4 (major)	CYP3A4 (major)	Covalent adducts	CYP2D6 (major)	CYP3A4 (major)	CYP3A (major)
Excretion	Feces (86%)	Feces (83%), Urine (8%)	Feces (85%), Urine (4%)	Feces (79%, intact 20%), Urine (3%)	Feces (68%), Urine (14%)	Feces (76%), Urine (4%)
Food effect	Not relevant	↑AUC 34-66%	High fat ↓AUC 39%	Not relevant	↑AUC 19%	Not relevant

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Adverse Effects: Small Molecule EGFR Inhibitors



Rash/acneiform dermatitis



Dry skin



Stomatitis



Diarrhea



Nausea/vomiting

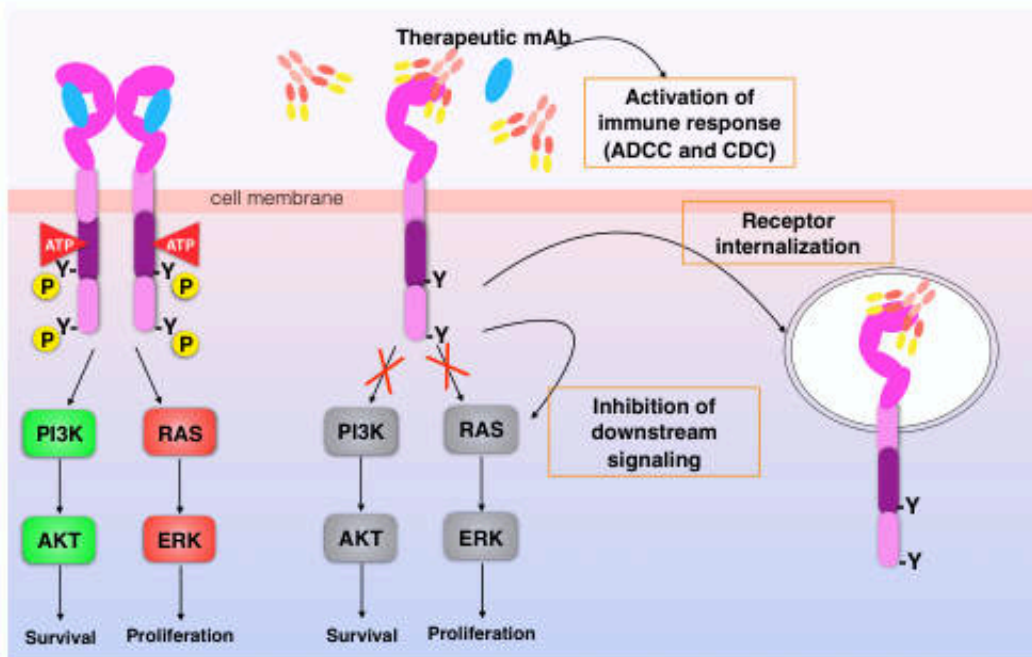


QT prolongation
(**mobocertinib**)

➡ Most AEs occur within the first month of therapy and are tolerable when managed with supportive medications and dose reductions

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EGFR Inhibitors: Monoclonal Antibodies (mAbs)



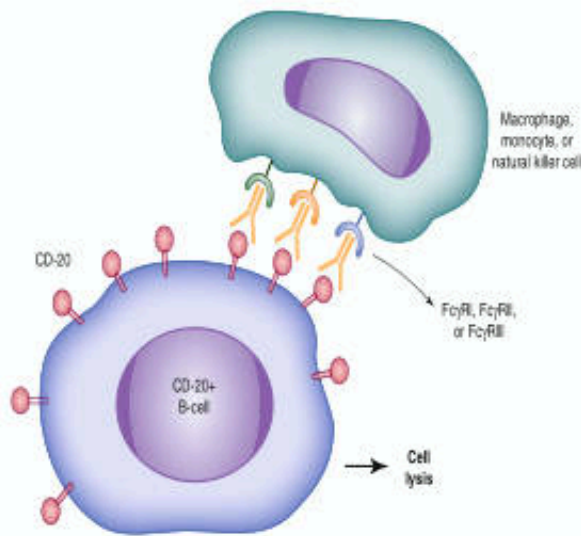
Mechanism of actions

- Preventing ligand-dependent signaling and receptor dimerization ➡ inhibition of EGFR phosphorylation
- Causing receptor internalization ➡ receptor down-regulation
- May also mediate ADCC against tumor cells (except panitumumab)

ADCC, Antibody-dependent cellular cytotoxicity; CDC, Complement-dependent cytotoxicity

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Antibody-Dependent Cellular Cytotoxicity (ADCC)

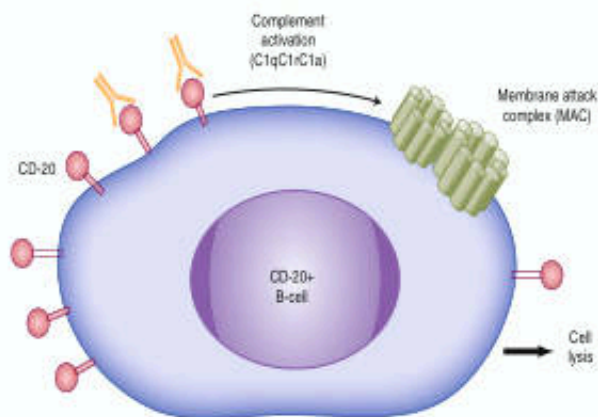


- ◆ ADCC is a mechanism of cell-mediated immunity
- ◆ An effector cell of the immune system: NK cells, monocytes, or macrophages
 - ⇒ *recognition of antibody-coated target cell by the Fc receptor*
 - ⇒ *releasing cytokines and cytotoxic granules*
 - ⇒ *triggering apoptosis*

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Pharmaceutical Biotechnology, 2013

Complement-Dependent Cytotoxicity (CDC)



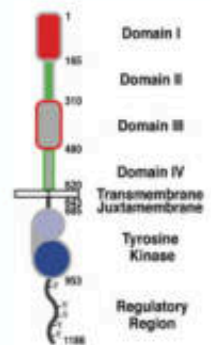
- ◆ CDC is initiated by binding of the complement protein, C1q, to the Fc domain
- ◆ IgG1 and IgG3 isotypes have the highest CDC activity
- ◆ IgG4 isotype lacks C1q binding and complement activation

Pharmaceutical Biotechnology, 2013

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EGFR Inhibitors: Monoclonal Antibodies (mAbs)

	Cetuximab (Erbix[®])	Panitumumab (Vectibix[®])	Necitumumab (Portrazza[®])
Structure	Chimeric IgG1 mAb	Recombinant human IgG2 mAb	Recombinant human IgG1 mAb
Binding sites in the EGFR-binding pocket	Domain III (D355, Q408, H409, K443, S468)	Domain III (D355, K443)	S468R variant of EGFR domain III (as an inhibitor of cetuximab- and/or panitumumab-resistant EGFR)
t1/2 (d), mean	~5	7.5	14
Metabolism	Proteolytic cleavage	Proteolytic cleavage	Proteolytic cleavage



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Adverse Effects of Anti-EGFR mAb



Infusion reaction

- Flushing, fever and chills
- In severe case: anaphylactic reactions, bronchospasm, and hypotension



Dermatotoxicity

- Acneiform rash
- Skin drying
- Skin exfoliation
- Skin fissure
- "sunlight can exacerbate dermatologic toxicity"



Cardiopulmonary arrest

- The cessation of adequate heart function and respiration

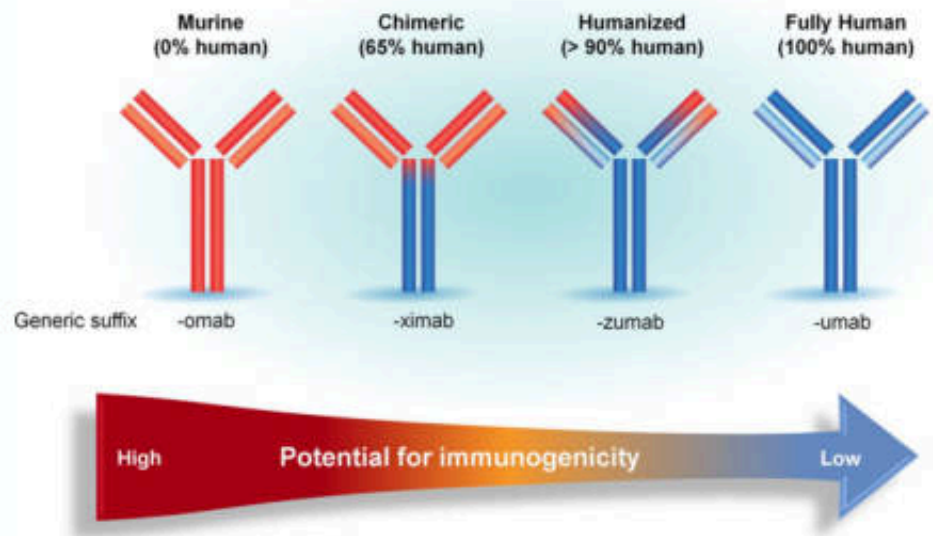
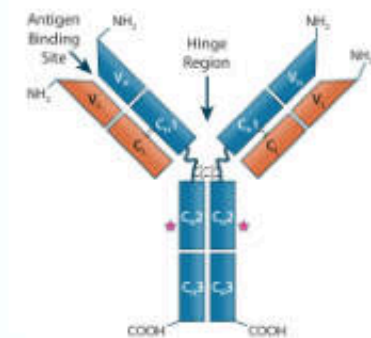


Hypomagnesemia

- Patients' electrolytes should be periodically monitored

20

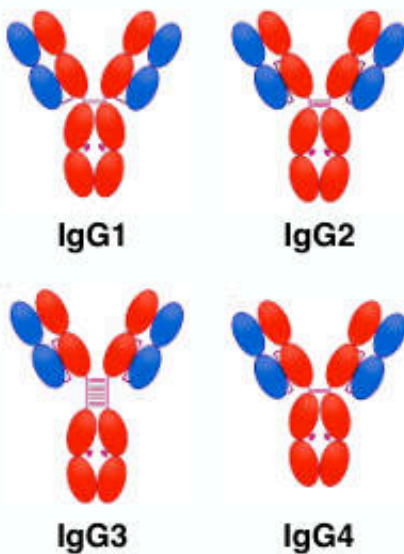
Different Generations of Therapeutic Monoclonal Antibodies



21

Circulation. 2013;127:2222-2230.

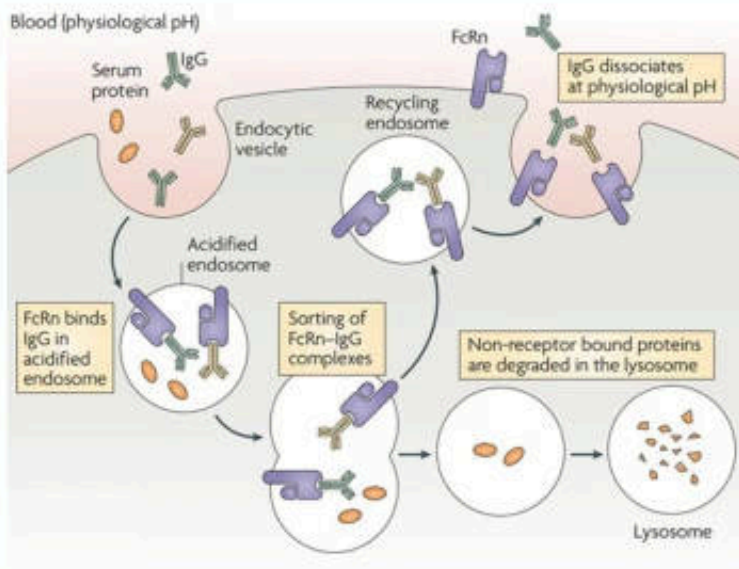
IgG Subclasses



	IgG1	IgG2	IgG3	IgG4
MW (kDa)	~150	~150	~160	~150
Heavy chain constant gene	γ ₁	γ ₂	γ ₃	γ ₄
No. of disulfide in hinge	2	4	11	2
Relative abundance	60%	32%	4%	4%
Serum half-life (days)	21-24	21-24	7-8	21-24
Complement activation	++	+	+++	-
Bind to Fc receptors	+++	+/-	++++	++
Cross placenta	+	+/-	+	+

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FcRn-mediated IgG recycling process

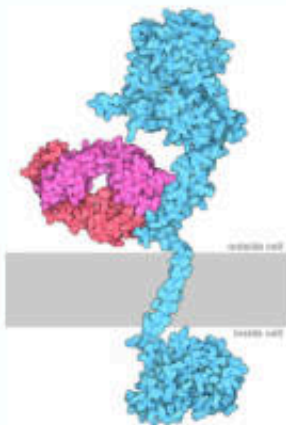


Mechanism of FcRn to prolong the half-life

- IgG has a half-life of up to 2 to 4 weeks in the human body
- FcRn binds IgG in acidic conditions (pH 6.0–6.5)
- Dissociation takes place under neutral and weakly alkaline conditions (pH 7.0–7.5)
- The IgG-FcRn complex is transferred to the cell surface through the recirculated endosome
- IgG1 inhibits FcRn-mediated transport and rescue of human IgG3, leading to lysosomal degradation of IgG3

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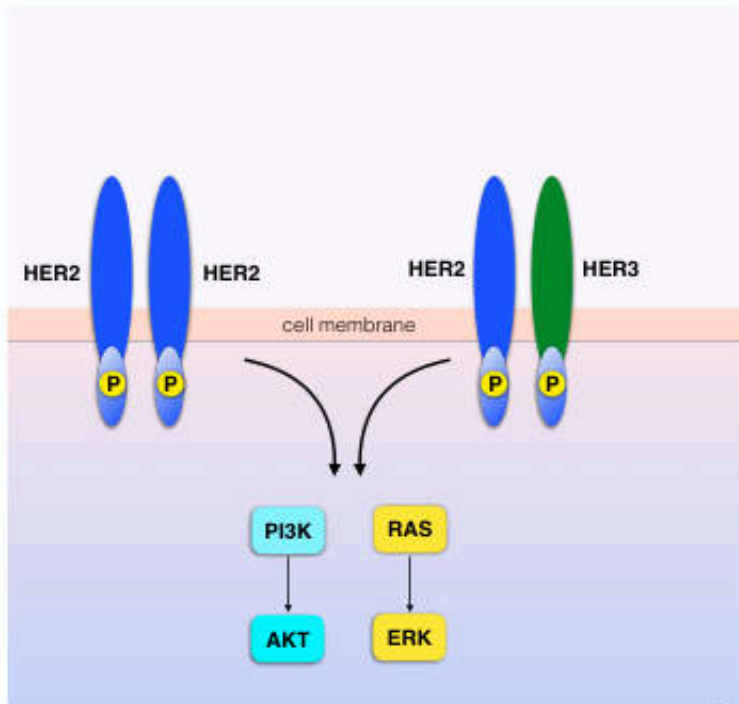
Nat Rev Immunol. 2007 Sep;7(9):715-25



HER2 Inhibitors

- ♦ *small molecule Inhibitors/tyrosine kinase inhibitors (TKIs)*
- ♦ *monoclonal antibodies (mAbs)*

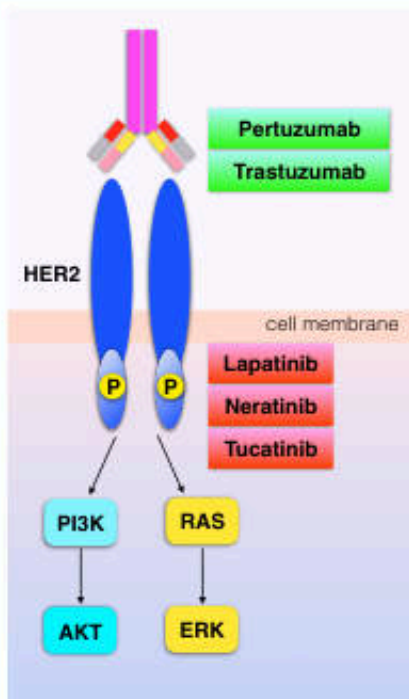
HER2 Inhibitors



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- ✦ HER2 amplification/overexpression is an effective therapeutic target in breast and gastric cancer
- ✦ Overexpression of HER2 in breast cancer leads to increased homodimerization (HER2:HER2) and heterodimerization (e.g., HER2:HER3), which initiates a strong pro-tumorigenic signaling cascade

HER2 Inhibitors

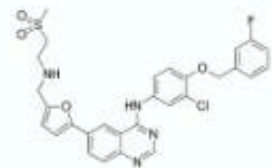


Monoclonal antibodies

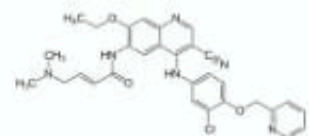
- inhibit the HER2 dimerization, preventing kinase activity and subsequent activation of signaling pathways that promote cancer cell growth

Small molecule inhibitors

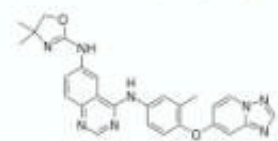
- inhibit the tyrosine kinase activity by binding to their ATP-binding pockets, preventing kinase activity and subsequent activation of signaling pathways that promote cancer cell growth



Lapatinib (Tykerb®)



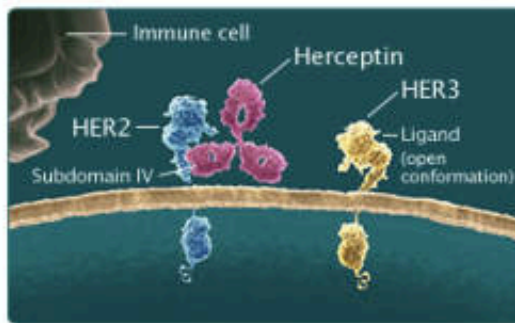
Neratinib (Nerlynx®)



Tucatinib (Tukysa®)

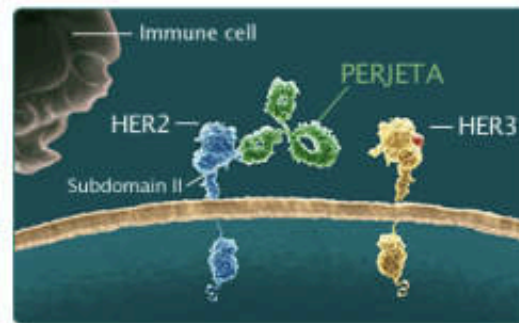
26

Anti-HER2 Monoclonal Antibodies



Trastuzumab (HERCEPTIN®):

- Inhibits **ligand-independent** HER2 signaling
- Activates ADCC
- Prevents HER2 extracellular domain (ECD) shedding



Pertuzumab (PERJETA®):

- Inhibits **ligand-dependent** HER2 dimerization and signaling
- Activates ADCC

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<http://www.perjeta.com/>

HER2 Inhibitors

Drug	Type	Route of Administration	Target	Metabolism	t1/2	Protein Binding (%)	Food Effect
Trastuzumab (Herceptin®)	mAb, IgG1	IV	HER2	Proteolytic cleavage	12 d (500 mg dose)		
Pertuzumab (Perjeta®)	mAb, IgG1	IV	HER2	Proteolytic cleavage	18 d		
Lapatinib (Tykerb®)	TKI, reversible	Oral	HER1, HER2	CYP3A4 (major)	24 h	>99	↑AUC 3-4 folds
Neratinib (Nerlynx®)	TKI, irreversible	Oral	HER1, HER2, HER4	CYP3A4 (major)	~14 h (~22 h active metabolite M3)	>99	↑AUC 1-2 folds
Tucatinib (Tykysa®)	TKI, reversible	Oral	HER2, HER3	CYP2C8 (major)	~8.5 h	97	↑AUC 1.5 folds

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Adverse Effects of HER2 Inhibitors



Diarrhea



Rash



Left ventricular dysfunction



Infusion reaction
(Trastuzumab,
pertuzumab)



Nausea/vomiting



Embryo-fetal toxicity



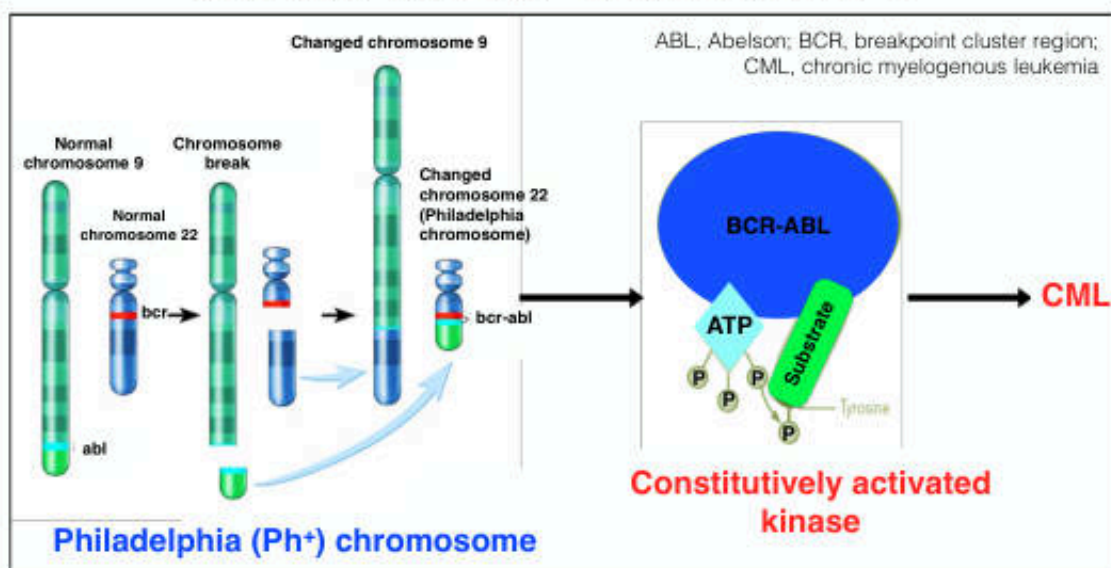
Hepatotoxicity



QT prolongation
(Lapatinib)

29

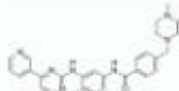
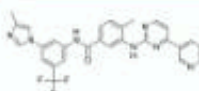
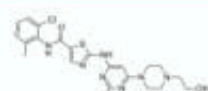
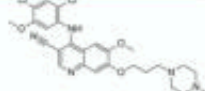
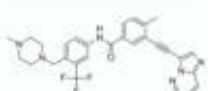
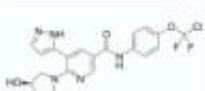
BCR-ABL Inhibitors



The most common BCR-ABL1 isoform results from the fusion of *BCR* exon 13 or 14 (e13/e14) with *ABL1* exon 2 (a2)

30

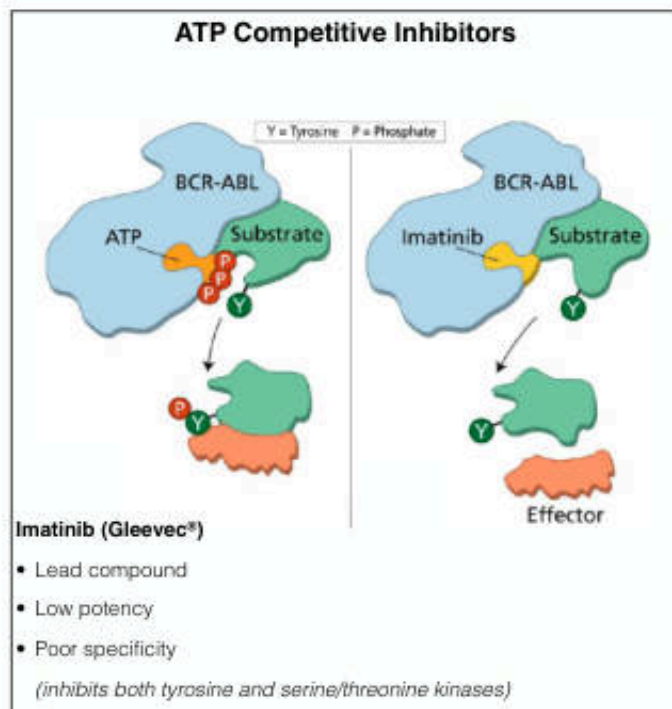
BCR-ABL Inhibitors

	1 st generation	2 nd generation		3 rd generation	4 th generation	
						
	Imatinib (Gleevec®)	Nilotinib (Tasigna®)	Dasatinib (Sprycel®)	Bosutinib (Bosulif®)	Ponatinib (Iclusig®)	Asciminib (Scemblix®)
Target	BCR-ABL, PDGF, SCF, c-kit	BCR-ABL, PDGFR, c-Kit, CSF-1R, DDR	BCR-ABL, SRC family, c-KIT, EPHA2, PDGFR	BCR-ABL, SRC family	BCR-ABL, VEGFR, PDGFR, FGFR, EPHR, SRC family, KIT, RET, TIE2, FLT3	ABL/BCR::ABL1
Binding conformation on BCR-ABL	Inactive	Inactive	Active	Both	Inactive	Myristoyl pocket
Resistance	Y253 D252 E255 F317 T315 M351 M244 M355 L248 F359 G250 H396	T315 L248 Y253 E255. F359	T315 V299 F317	T315 V299 L248 G250 E255 F317	T315 E255	A337 W464 P465 V468 I502

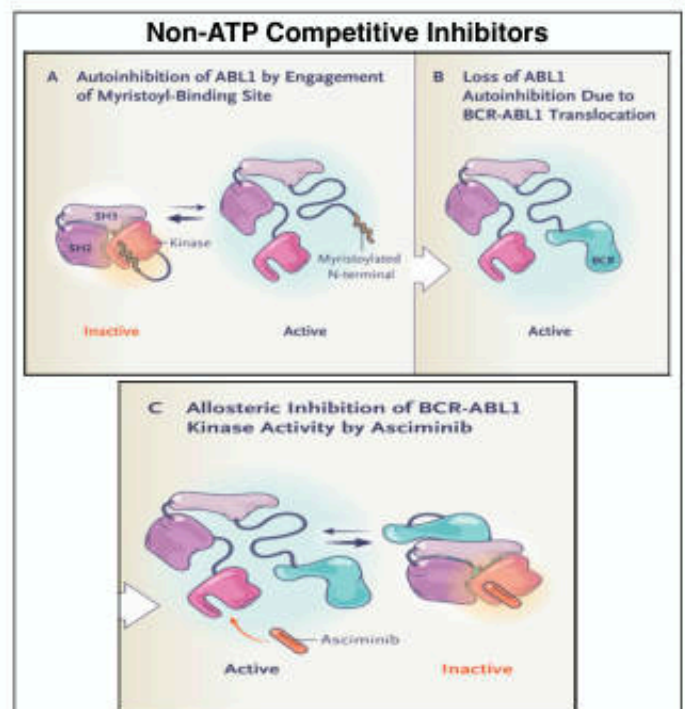
- Residues at which mutations are associated with strong resistance to a given TKI are indicated in red
- Both T315 and E255 mutations do lead to an increase in the IC50 for ponatinib; however, they do not typically lead to clinical resistance in isolation

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Mechanisms of BCR-ABL Inhibitors



Front Biosci. 2006;11:209-20



32

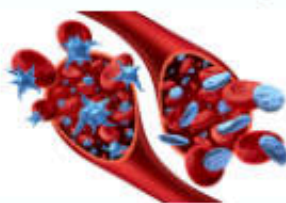
N Engl J Med. 2019;381(24):2315-2326

BCR-ABL Inhibitors

	Imatinib (Gleevec®)	Nilotinib (Tasigna®)	Dasatinib (Sprycel®)	Bosutinib (Bosulif®)	Ponatinib (Iclusig®)	Asciminib (Scemblix®)
F	~98%	17-44%	Not known	~33%	Not known	~73%
Protein binding	~95%	~98%	~96%	~96%	~99%	97.3%
Metabolism	CYP3A4	Oxidation and hydroxylation	CYP3A4 (major)	CYP3A4 (major)	CYP3A4 (major)	CYP3A4, UGT2B7, UGT2B17
t1/2	18 h (40 h for active metabolite)	~17 h	3-5 h	~22.5 h	~24 h	5.2 h
Elimination	Feces (~68%)	Feces (~93%)	Feces (~85%)	Feces (~91%)	Feces (87%)	Feces (with 56.7% intact)
Food effects	No relevant	↑AUC 82% (high-fat meal)	↑AUC 14% (high-fat meal)	↑AUC 1.7 folds (high-fat meal)	No relevant	↓AUC 62.3% (high-fat meal)

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Adverse Effects of BCR-ABL Inhibitors



Myelosuppression

- anemia
- neutropenia
- thrombocytopenia



GI distress

- nausea/vomiting
- abdominal pain
- diarrhea



Hepatotoxicity



Edema and fluid retention



Rash



Cardiotoxicity

- left ventricular dysfunction
- congestive heart failure
- cardiogenic shock



Embryo-fetal toxicity

- teratogenic effect
- early fetal resorption
- nonviable pups



QT prolongation (nilotinib, dasatinib, asciminib)



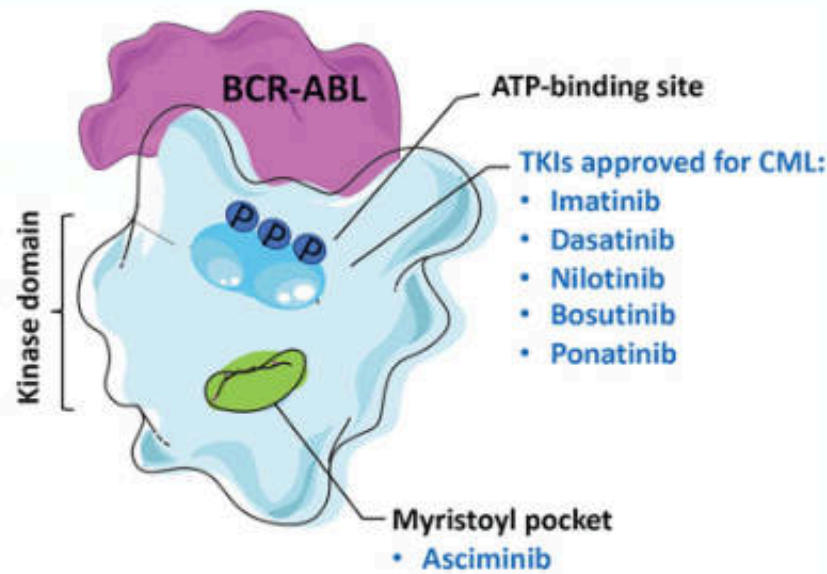
Thrombosis (ponatinib)



Hypertension (ponatinib, asciminib)

34

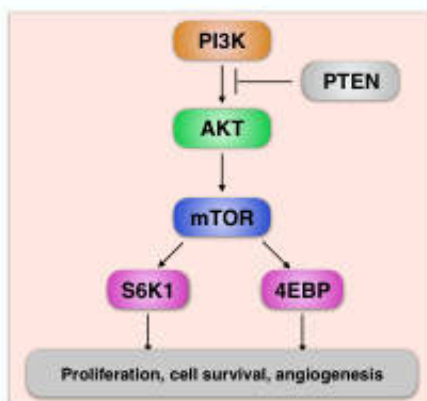
BCR-ABL Inhibitors



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Front Oncol. 2021 Dec 15;11:801779

mTOR Inhibitors



- ✦ **Mammalian Target Of Rapamycin (mTOR)** is a serine/threonine protein kinase
- ✦ Many human tumors occur because of dysregulation of mTOR signaling (e.g., loss of PTEN function)
- ✦ mTOR inhibitors found important applications in the treatment of renal and hepatocellular cancer and mantle cell lymphoma

S6K1, Ribosomal protein S6 kinase beta-1
4EBP, Eukaryotic initiation factor 4e binding protein

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mTOR Inhibitors

- Antitumor activity of mTOR inhibitors results from their binding to **FKBP12** and inhibition of **mTORC1**

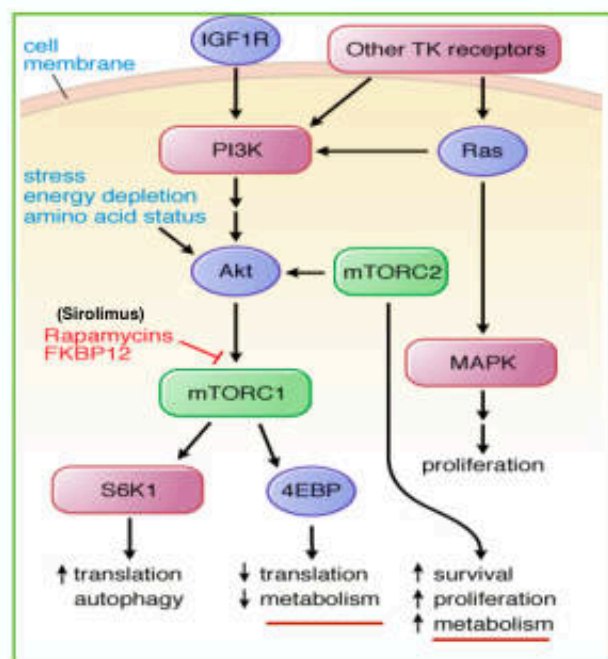
- ⇒ **G₁/S phase block (inhibit cell cycle progression)**
- ⇒ **Inhibit angiogenesis**
- ⇒ **Promote apoptosis**

- Resistance to mTOR inhibitors is incompletely understood

- ⇒ **May arise through the action of mTORC2**

Note: Tacrolimus (FK506) binds to FKBP12 and inhibit calcineulin

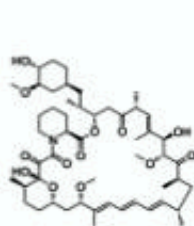
4EBP, Eukaryotic initiation factor 4e binding protein



37

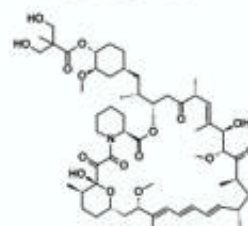
Goodman and Gilman's: The Pharmacological Basis of Therapeutics, 12th Ed

mTOR Inhibitors



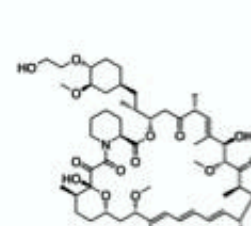
Sirolimus (Rapamycin)
(Rapamune®)

IV and oral



Temsirolimus
(Torisel®)

IV



Everolimus
(Afinitor®)

Oral

Route of administration

Metabolism

CYP3A4 and substrate of P-gp

CYP3A4

CYP3A4 and substrate of P-gp

t_{1/2}

~60 h

~17 h

Note

Oral dosage form should be taken consistently with or without food (high-fat meal ↑AUC 23-35%)

- high-fat meals reduced systemic exposure by 22%
- However, food had no apparent effect on the post absorption phase concentration-time profile

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Adverse Effects of mTOR Inhibitors



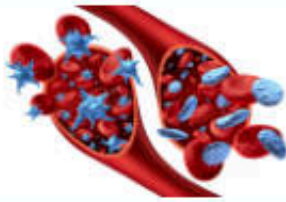
Oral ulceration



Hyperlipidemia



Hyperglycemia



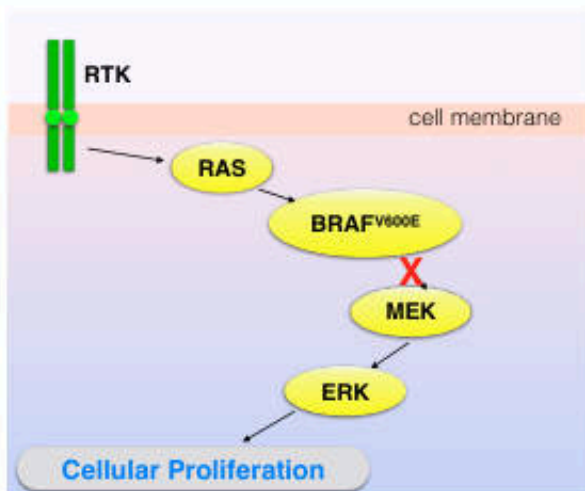
Anemia
Thrombocytopenia



Edema

39

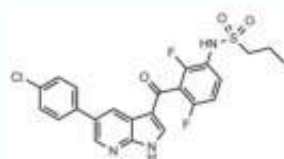
BRAF Inhibitors



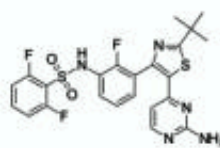
- ❖ BRAF is a component of the MAP kinase pathway
- ❖ BRAF mutations are found in melanoma, colorectal cancer, non-small cell lung cancer, and multiple myeloma

40

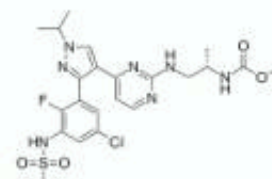
BRAF Inhibitors



Vemurafenib
(Zelboraf®)



Dabrafenib
(Tafinlar®)



Encorafenib
(Braftovi®)

Activity

BRAF^{V600E}, V600D, V600R mutant
CRAF, ARAF, wild-type BRAF,
SRMS, ACK1, MAP4K5, FGR

BRAF^{V600E}, V600D, V600R, V600K mutant
wild-type BRAF, CRAF

BRAF^{V600E}, V600D, V600K mutant
wild-type BRAF, CRAF

Metabolism

CYP3A4

CYP2C8 and CYP3A4

CYP3A4

t_{1/2}

~50 h

~8 h

~3.5 h

Note

high-fat meal ↑AUC 5 folds

high-fat meal ↓AUC 5 folds

high-fat meal ↓C_{max} by 36%
with no effect on AUC

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Adverse Effects of BRAF Inhibitors



Dermatotoxicity

- rash
- skin papilloma
- hyperkeratosis
- photosensitivity



Alopecia



Hepatotoxicity



Embryo-fetal toxicity



QT prolongation (vemurafenib, encorafenib)

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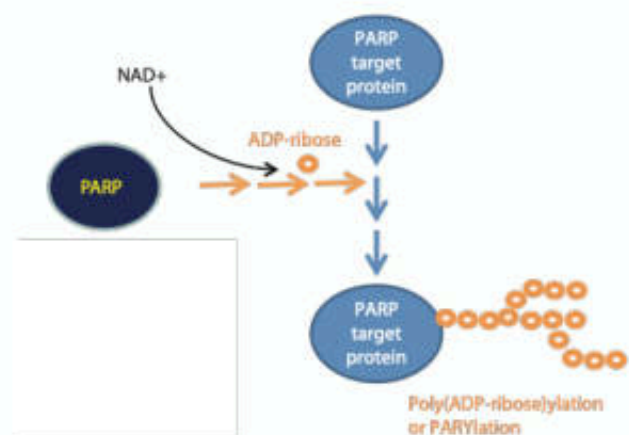
PARP Inhibitors

PARP, Poly(ADP-Ribose) Polymerases (PARPs)

43

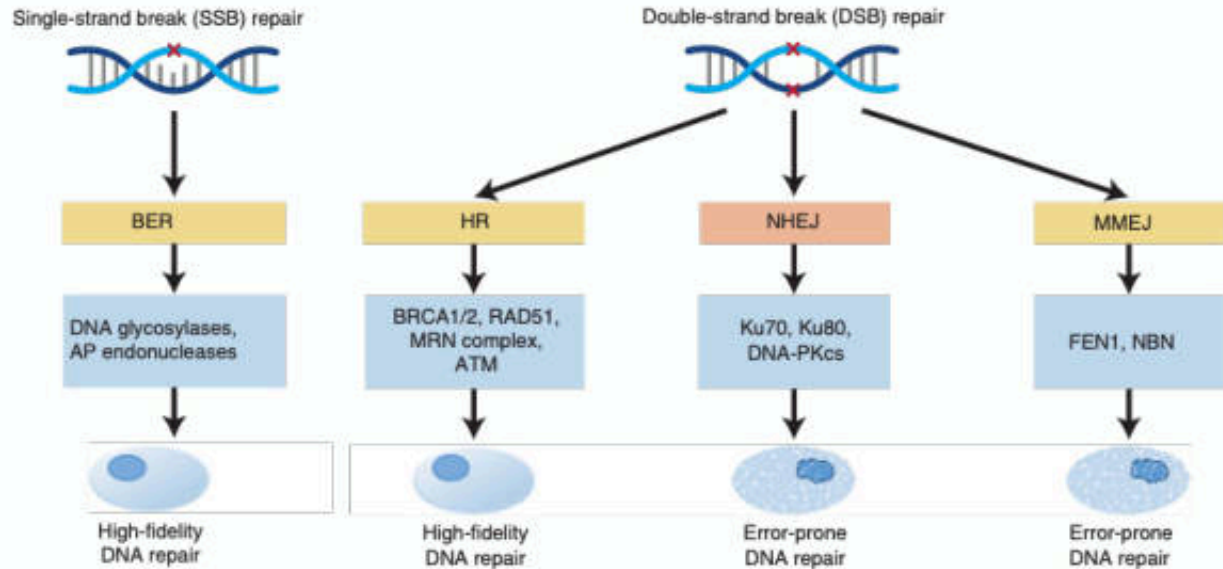
Poly(ADP-Ribose) Polymerases (PARPs)

- A family of enzymes involved in cell signaling for DNA damage response
- Transferring ADP-ribose moieties from the dinucleotide nicotinamide adenine dinucleotide (NAD⁺) to certain polypeptides resulting in mono- or poly(ADP-ribosylation) of these substrates



44

Main DNA Repair Mechanisms

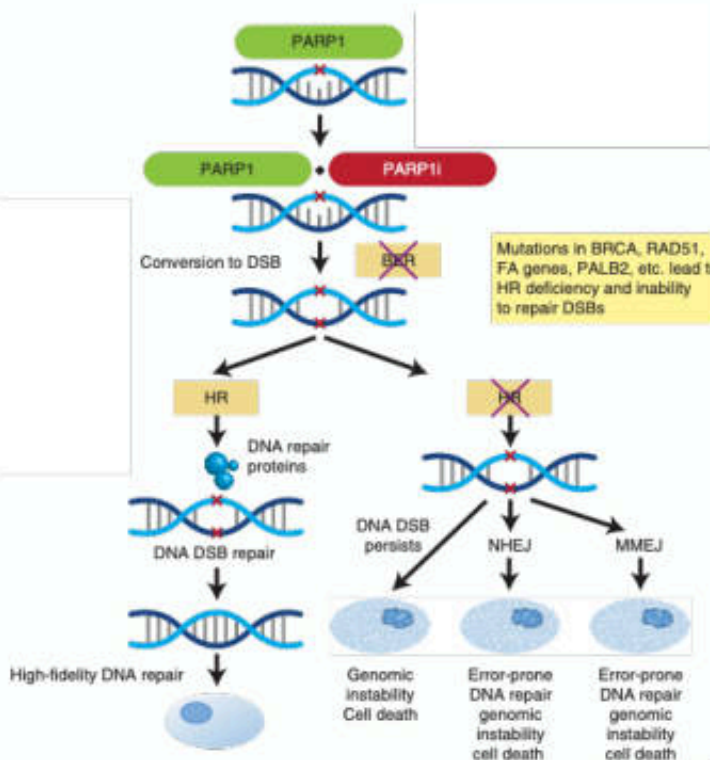


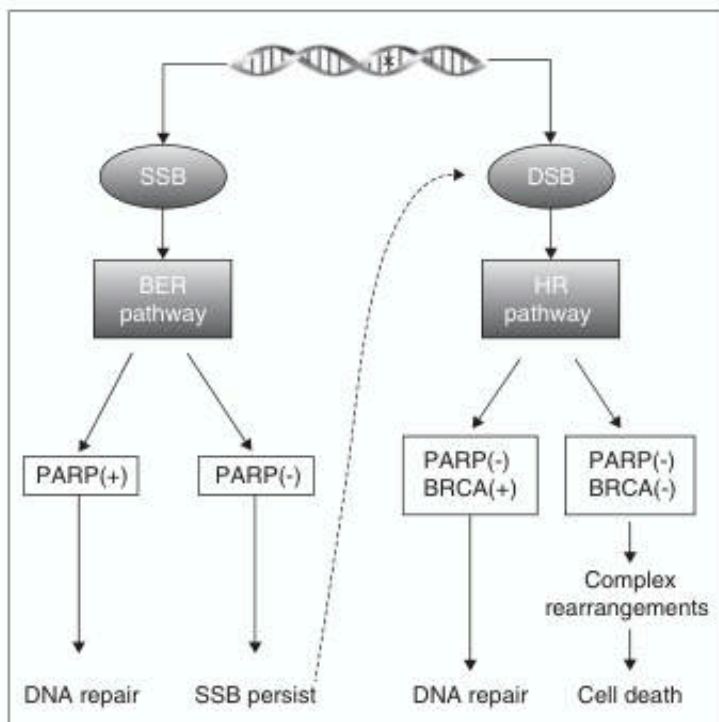
BER, base-excision repair; HR, homologous recombination; MMEJ, microhomology-mediated end joining; NHEJ, non-homologous end joining

Br J Cancer. 2016 Nov 8;115(10):1157-1173.

Synthetic Lethality Between PARP and BRCA1/2

- PARP inhibitors cause an increase in DNA single strand breaks (SSBs), if left unrepaired, will lead to double strand breaks (DSBs)
 - using homologous recombination (HR) repair
- Inhibition of PARP was shown to be highly selective for cancer cells that harbor HR deficiencies
 - such as mutations in *BRCA1* or *BRCA2*



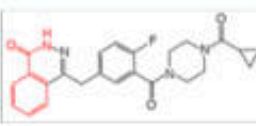
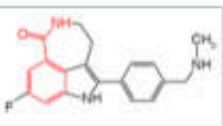
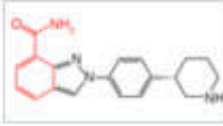
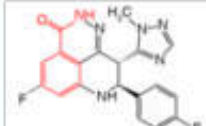


The synthetic lethal interaction between PARP and BRCA has been proposed as a potential explanation for the **sensitivity of BRCA mutation** cell lines to PARP inhibition

Expert Opin Investig Drugs (2009) 18(1):31-43

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PARP Inhibitors

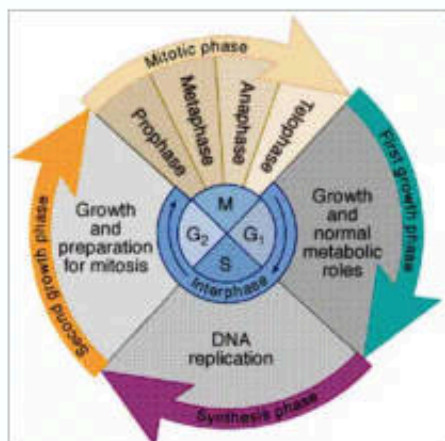
Agents	 Olaparib	 Rucaparib	 Niraparib	 Talazoparib
Target	PARP1, PARP2, PARP3	PARP1, PARP2, PARP3	PARP1, PARP2	PARP1, PARP2
PARP1 IC ₅₀	5 nM	3.8 nM	2	0.57
PARP Trapping Potency (Relative to Olaparib)	1	1	2	100
Company	Astra Zeneca	Clovis Oncology	Tesaro	Pfizer

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PARP Inhibitors: Adverse Effects

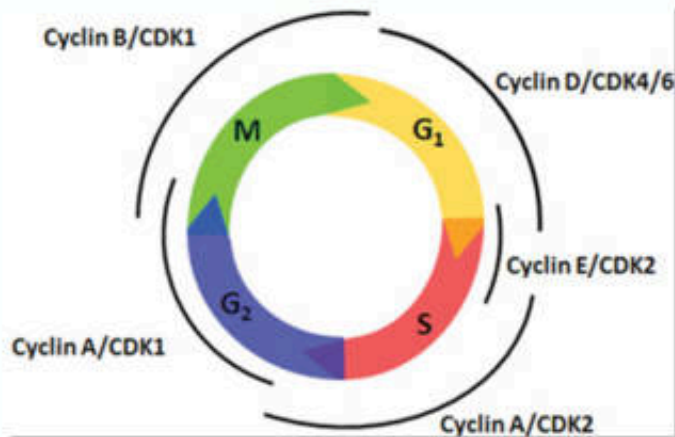
Agents	Olaparib	Rucaparib	Niraparib	Talazoparib
Adverse effects (most frequent)	Nausea Fatigue Vomiting Diarrhea Dysgeusia Headache	Nausea Fatigue Vomiting Diarrhea Dysgeusia LFT elevation	Nausea Fatigue LFT elevation Vomiting Headache Insomnia Hypertension	Nausea Fatigue Headache Vomiting Alopecia Diarrhea
Serious hematologic toxicities (Grade ≥ 3 in $\geq 5\%$ of study population)	Anemia Neutropenia	Anemia Neutropenia	Thrombocytopenia Anemia Neutropenia	Anemia Neutropenia Thrombocytopenia

PARP inhibitors have been shown to cause teratogenicity, and embryo-fetal toxicity and death in animal reproduction studies, and should, therefore, be avoided during pregnancy



CDK4/6 Inhibitors

The Cell Cycle



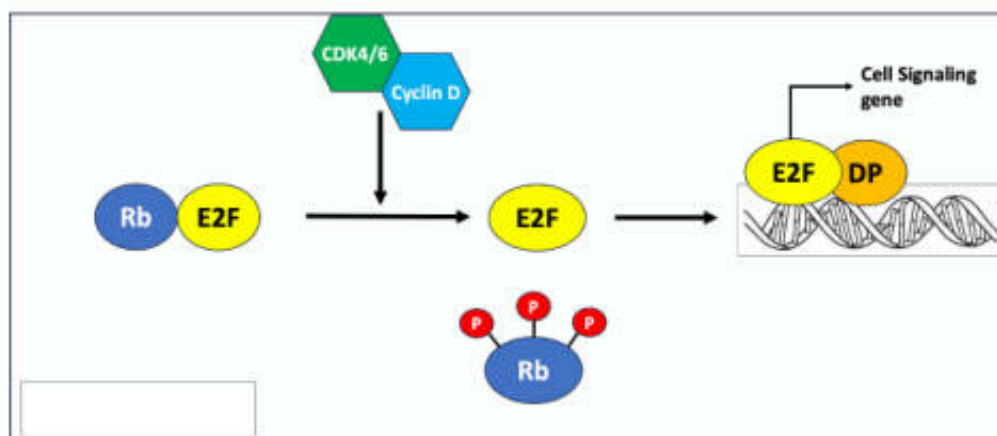
- The mammalian cell cycle consists of
 - Gap 1 (G₁) phase
 - DNA synthesis (S) phase
 - Gap 2 (G₂) phase
 - Mitotic (M) phase
- Different cyclin-CDK complexes regulate progression of cells through the different phases of the cell cycle

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Biosci Rep. 2010 Mar 17;30(4):243-55.

Cyclin D-CDK4/6 and Cell Cycle Progression

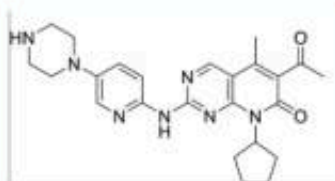
- Cyclin D-CDK4/6 in driving cell proliferation is phosphorylation of the retinoblastoma protein (Rb)



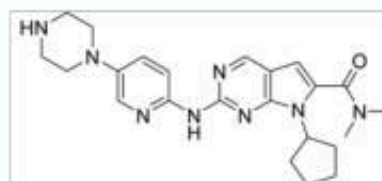
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CDK4/6 Inhibitors

IC50
CDK4: 9-11 nM
CDK6: 15 nM
CDK1: >10 uM
CDK2: >10 uM

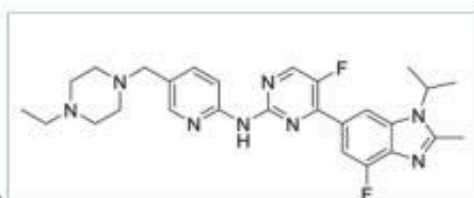


Palbociclib (Ibrance®)



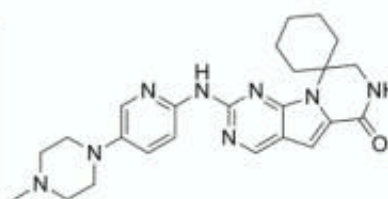
Ribociclib (Kisqali®)

IC50
CDK4: 10 nM
CDK6: 39 nM
CDK1: >100 uM
CDK2: >50 uM



Abemaciclib (Verzenio®)

IC50
CDK4: 2 nM
CDK6: 5 nM
CDK1: >1 uM
CDK2: >500 nM



Trilaciclib (Cosela®)

IC50
CDK4: 1 nM
CDK6: 4 nM
CDK1: >1 uM
CDK2: >2 uM

Trilaciclib indication: to decrease the incidence of chemotherapy-induced myelosuppression in adult patients when administered prior to a platinum/etoposide-containing regimen or topotecan-containing regimen for extensive-stage small cell lung cancer

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CDK4/6 Inhibitors

	Palbociclib	Ribociclib	Abemaciclib
Mechanism	ATP-competitive, reversible, selective CDK4/6 inhibitor		
Targets	CDK4 and CDK6	CDK4 and CDK6	CDK4 and CDK6 (have been shown to have <i>in vivo</i> inhibition of CDK1, CDK2, CDK5, CDK9, CDK14)
Cell cycle arrest	G1 phase	G1 phase	G1 and G2 phase
CNS activity	-	-	+
Dosing	125 mg once daily for 21 days followed by 7 days off	600 mg once daily for 21 days	150 mg twice daily

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CDK4/6 Inhibitors: Pharmacokinetics

	Palbociclib	Ribociclib	Abemaciclib
Bioavailability	46%	NA	45%
Protein binding	~85%	~70%	96%
Vd	2583 L	1090 L	750L
Metabolism	CYP3A and SULT2A1	CYP3A4	CYP3A4
Half-life	~29 h	~32 h	~25 h
Excretion	Feces 74% (parent 2%), Urine 17% (parent 7%)	Feces 69% (parent 14%), Urine 23% (parent 4%)	Feces 81%, Urine 3.4%
Food effect	High fat ↑ AUC 22%	No effect	High fat ↑ AUC 9%

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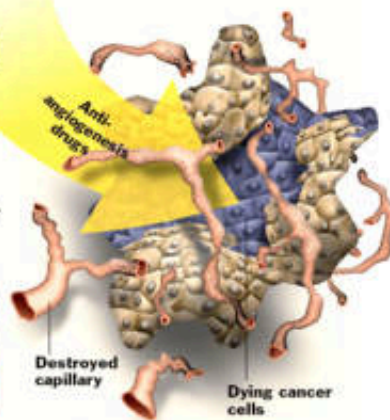
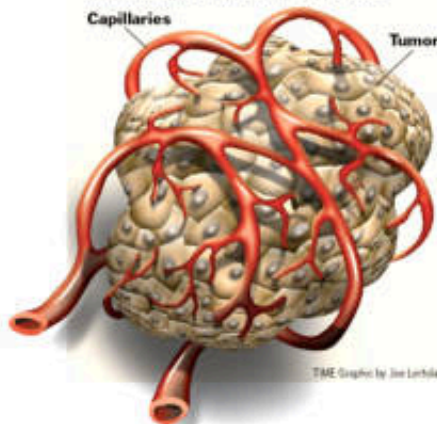
CDK4/6 Inhibitors: Adverse Effects

	Palbociclib	Ribociclib	Abemaciclib
Common adverse effects	neutropenia, leukopenia and fatigue	neutropenia, increased creatinine, hyponatremia, QTc prolongation	diarrhea and fatigue
Myelosuppression	++	++	+
GI toxicity	+	+	++
Pneumonitis	+ (rare but serious)	+ (rare but serious)	+ (rare but serious)
LFT abnormalities	-	+	+
QTc prolongation	-	+	-

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HOW TO STARVE A TUMOR

1 As a cancer tumor grows, it builds its own network of capillaries that tap into the body's blood supply and draw on the oxygen and nutrients the tumor needs to survive

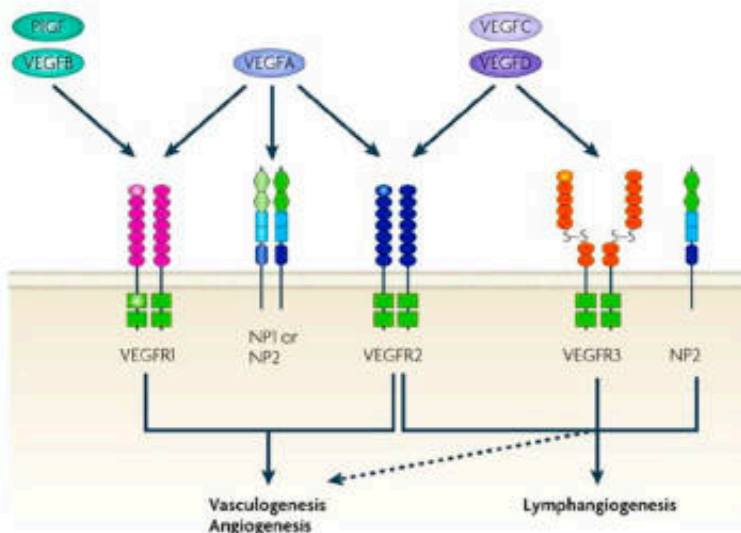


2 Drugs that block angiogenesis—the formation of new vessels—cut the tumor off from its blood supply. Gradually, malignant cells die and the tumor starts to shrink

Angiogenesis Inhibitors

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VEGF Signaling Pathway



Nat Rev Cancer. 2008 Aug;8(8):579-91.

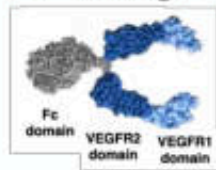
Nature Reviews | Cancer

- VEGF-A and its receptors VEGFR-1 and VEGFR-2 play major roles in physiological and pathological angiogenesis
- **Affinity** of VEGFR-1 for VEGF-A ($K_d = 1\sim 10\text{ pM}$) > VEGFR-2
- **Tyrosine kinase activity** of VEGFR-1 is $\sim 10\text{-fold}$ < VEGFR-2

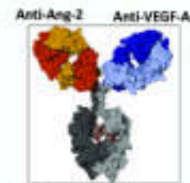
VEGF Inhibitors: Targeting VEGF-A



Bevacizumab (Avastin®)



ziv-Aflibercept (Zaltrap®)



Faricimab (Vabysmo®)

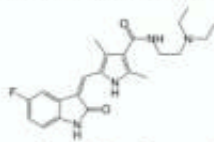
Type	A humanized antibody IgG1 against VEGF-A	A recombinant fusion protein consisting of portions of human VEGF receptors 1 and 2 extracellular domains fused to the Fc portion of human IgG1	A bispecific IgG1 antibody against angiopoietin 2 (Ang-2) and VEGF-A (Modification of Fc domain disabled the antibodies' Fc-mediated effector functions)
t1/2	~20 days	~5-6 days	~7.5 days
Indication	Various types of cancer	- Wet age-related macular degeneration (AMD) - Diabetic macular edema (DME)	- AMD - DME
Note	It neutralising the biological activity of VEGF → inhibiting vascularization of tumor and thereby inhibiting tumor growth	It is produced in Chinese hamster ovary (CHO) K1 cells by recombinant DNA technology	It reduces vascular permeability and inflammation, resulting in restoration of vascular stability

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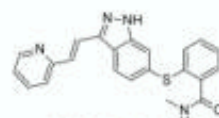
VEGF Inhibitors: Targeting VEGFR



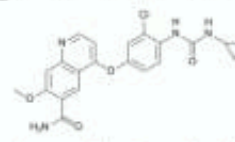
Ramucirumab (Cyramza®)



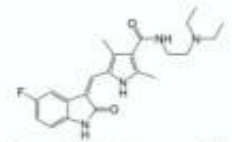
Sunitinib (Sutent®)



Axitinib (Inlyta®)



lenvatinib (Lenvima®)



Sorafenib (Nexavar®)

Type	A recombinant human IgG1 antibody against VEGFR-2	A multikinase reversible, ATP-competitive inhibitor : VEGFR1-3, PDGFRα, PDGFRβ, KIT, FLT3, RET, and CSF-1R	A reversible, ATP-competitive inhibitor : VEGFR1-3	A reversible, ATP-competitive inhibitor : VEGFR1-3, FGFR1-4, PDGFRα, KIT, and RET	A multikinase reversible, ATP-competitive inhibitor : c-CRAF, BRAF, mutant BRAF, KIT, FLT-3, RET, RET/PTC, VEGFR1-3, PDGFR-β
Protein binding	Not available	~95%	>99%	97-99%	99.5%
Metabolism	Not available (Proteolytic cleavage)	CYP3A4 (major)	CYP3A4 (major)	CYP3A (major)	CYP3A4 (major)
t1/2	~14 days	40-60 h	2.5-6 h	~28 h	40-60 h
Elimination	Not available	Feces (major)	Feces (major)	Feces (major)	Feces (major)
Food effects	No relevant	No relevant	No relevant	No relevant	↓ Bioavailability 29% (high fat meal)

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Adverse Effects of VEGF Inhibitors



Nausea/vomiting



Diarrhea



Abdominal pain



Fatigue



Vessel injury and bleeding

- GI perforation
- colonic perforation



Hypertension

: may ↓NO production



Arterial thrombosis

- : stroke
- : myocardial infarction



QT prolongation

(sunitinib, lenvatinib, sorafenib)

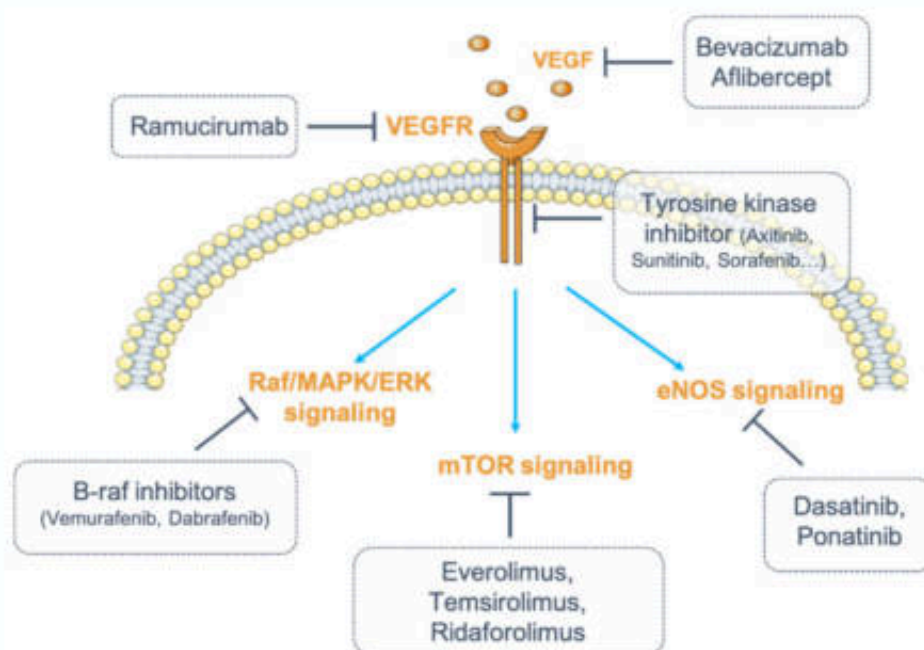


Hepatotoxicity

(sunitinib)

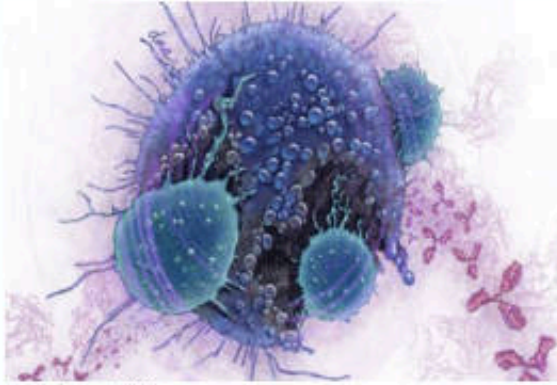
61

Angiogenesis Inhibitors



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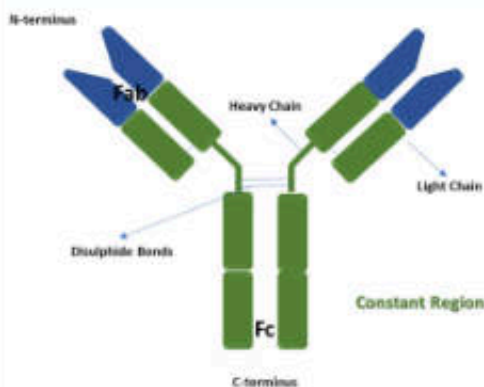
Immunotherapy



Br J Cancer. 2018

- Monoclonal antibodies
- Thalidomide and lenalidomide
- Interleukin-2
- Cancer vaccine
- Immunocheckpoint inhibitors
- G-CSF

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Monoclonal Antibodies (mAbs)

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Mechanism of Tumor Cell Killing by Antibodies

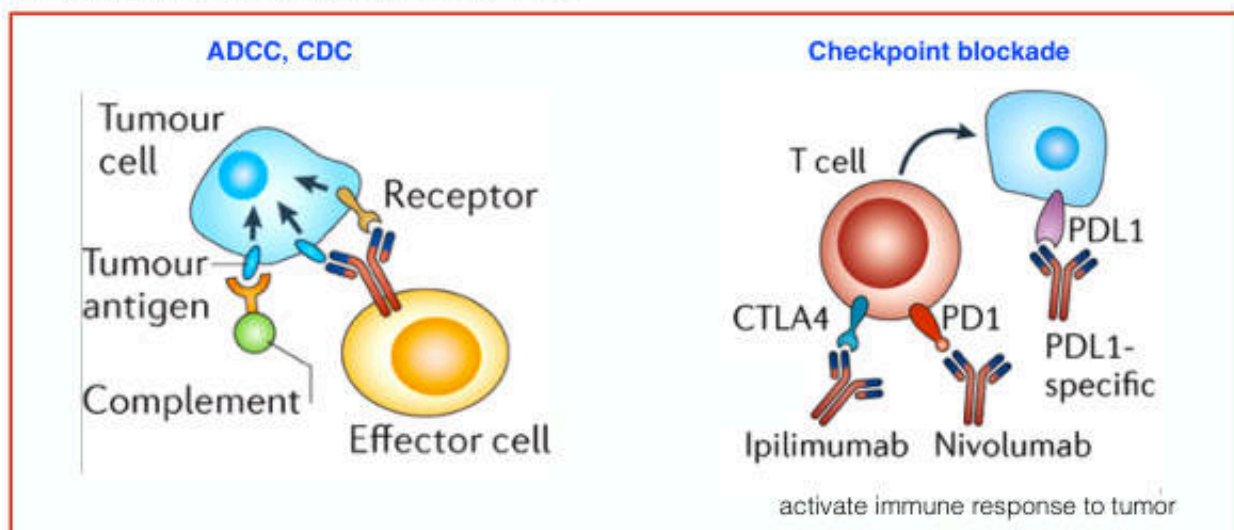
Direct Tumor Killing



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Mechanism of Tumor Cell Killing by Antibodies

Immune-Mediated Tumor Cell Killing



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Tumor Antigens as Targets for Antibody Therapy

Antigen category	Examples of antigens	Tumor types expressing antigen
Cluster of differentiation (CD) antigens	CD20	non-Hodgkin lymphoma
	CD30	Hodgkin lymphoma
	CD33	Acute myelogenous leukemia
	CD52	Chronic lymphocytic leukemia
Glycoproteins	EpCAM	Epithelial tumors (breast, colon, lung)
	CEA	Epithelial tumors (breast, colon, lung)
	gpA33	Colorectal carcinoma
	Mucins	Epithelial tumors (breast, colon, lung, ovarian)
	TAG-72	Epithelial tumors (breast, colon, lung)
	Carbonic anhydrase IX	Renal cell carcinoma
	PSMA	Prostate carcinoma
Glycolipids	Folate binding protein	Ovarian tumors
	Gangliosides (e.g., GD2, GD3, GM2)	Neuroectodermal tumors, some epithelial tumors
Carbohydrates	Lewis-Y ²	Epithelial tumors (breast, colon, lung, prostate)
Vascular targets	VEGF	Tumor vasculature
	VEGFR	Epithelium-derived solid tumors
	$\alpha V\beta 3$	Tumor vasculature
	$\alpha 5\beta 1$	Tumor vasculature
Growth factors	ErbB1/EGFR	Glioma, lung, breast, colon, head and neck tumors
	ErbB2/HER2	Breast, colon, lung, ovarian, prostate tumors
	ErbB3	Breast, colon, lung, ovarian, prostate tumors
	c-MET	Epithelial tumors (breast, ovary, lung)
	IGF1R	Lung, breast, head and neck, prostate, thyroid, glioma
	EphA3	Lung, kidney, colon, melanoma, glioma, hematological malignancies
	TRAIL-R1, TRAIL-R2	Solid tumors (colon, lung, pancreas) and hematological malignancies
	RANKL	Prostate cancer and bone metastases
	FAP	Epithelial tumors (colon, breast, lung, head and neck, pancreas)
Stromal and extracellular matrix antigens	Tenascin	Glioma, epithelial tumors (breast, prostate)

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Cancer Immunity (1 May 2012) Vol. 12, p. 14

mAbs: Adverse Effects

♦ Common adverse effects

- Allergic reactions, such as hives or itching
- Flu-like signs and symptoms, including chills, fatigue, fever, and muscle aches and pains
- Nausea
- Diarrhea
- Skin rashes

♦ Serious adverse effects

- **Infusion reactions:** severe allergy-like reactions ⇒ premedication to block an allergic reaction
- Certain monoclonal antibodies may cause : Heart problems, bleeding, cytopenia, skin problems



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Vaccine



Preventive Vaccine

- ✦ Prophylaxis
- ✦ Given before a person gets a disease to defend against infection
- ✦ Carrying immunologic specificity for individual infective disease agents and provoking an immune response against them
- ✦ Induction of antibody production

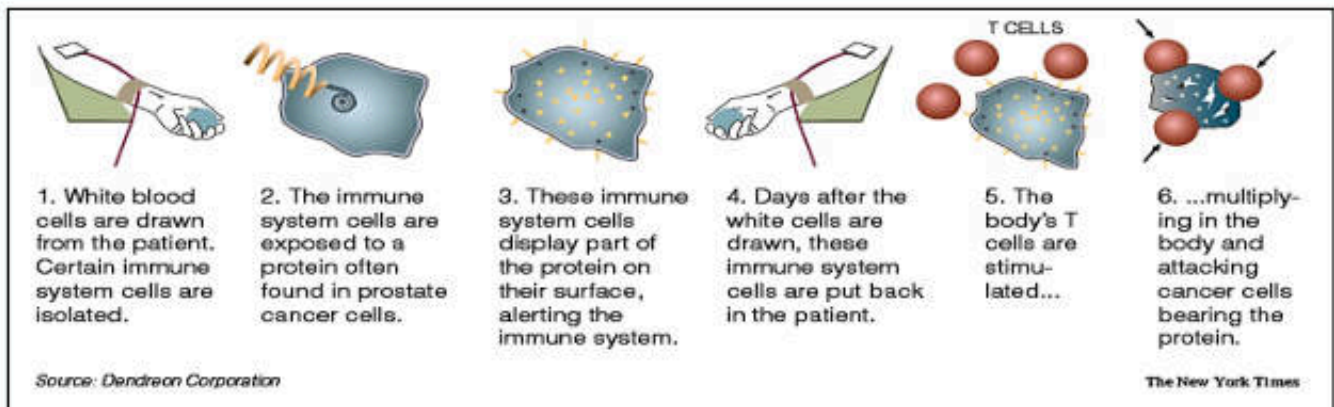
Therapeutic Vaccine

- ✦ Used for treatment
- ✦ Given after a person gets a disease
- ✦ Representing a viable option for active immunotherapy
- ✦ Two different approaches
 - Autologous vaccine (personalized vaccine)
 - Allogenic vaccine

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Cancer Therapeutic Vaccine

- ✦ The prostate cancer personalized vaccine, sipuleucel-T (Provenge®)
- ✦ Antigen: prostatic acid phosphatase (PAP)

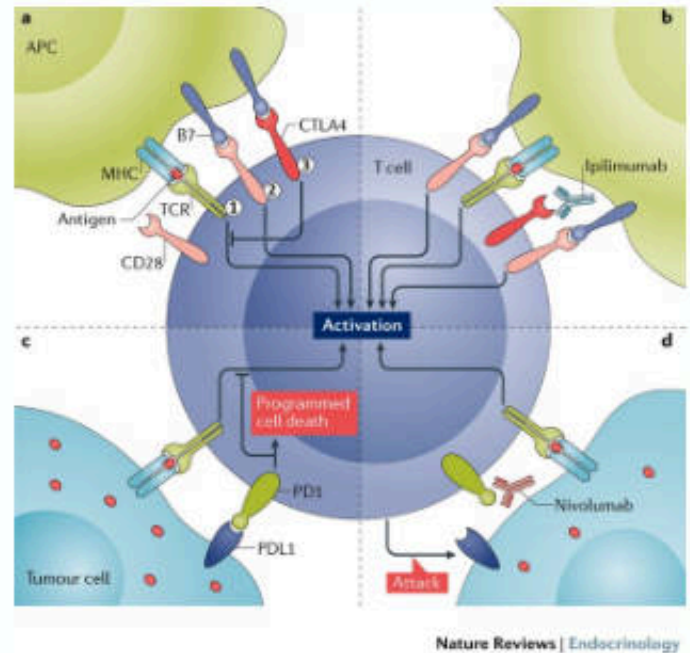


Adverse effects

- Chills
- Fatigue
- Fever
- Joint ache
- Back pain
- Nausea
- Headache

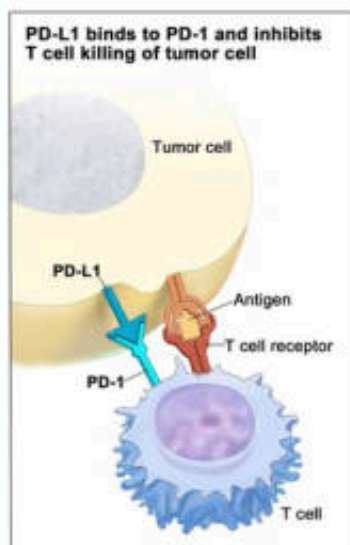
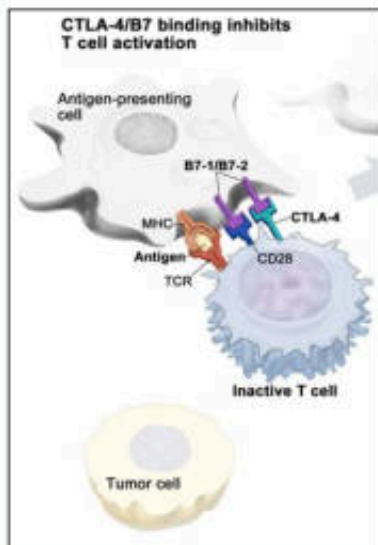
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Immune Checkpoint Inhibitors



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Immune Checkpoint Inhibitors



Immune Checkpoints

- ✦ Checkpoint proteins help keeping the body's immune responses in check
 - : **PD-L1** on tumor cells and **PD-1** on T cells
 - : **B7-1/B7-2** on antigen-presenting cells (APC) and **CTLA-4** on T cells
- ✦ They are a set of inhibitory and stimulatory pathways that directly affect the function of immune cells
- ✦ Cancer cells promote immunosuppressive state that favors immune evasion and tumor growth

PD-1, program cell death 1; CTLA-4, cytotoxic T lymphocyte antigen 4⁷²

The diagram illustrates the mechanism of action of immune checkpoint inhibitors (ICIs) in cancer immunotherapy. It shows a T cell interacting with a Dendritic cell and a Cancer cell. The T cell's TCR (T Cell Receptor) recognizes a peptide presented by the MHC (Major Histocompatibility Complex) on the Dendritic cell. The T cell also interacts with the Cancer cell via PD-1 (Programmed Death-1) and PD-L1 (Programmed Death-Ligand 1). The Dendritic cell expresses CTLA-4 (Cytotoxic T-Lymphocyte Antigen 4) and CD28 (Cluster of Differentiation 28). The Cancer cell expresses PD-L1 and B7 (Biomolecule 7). The diagram shows that anti-PD-1 antibodies (Pembrolizumab, Nivolumab, Cemiplimab) and anti-CTLA-4 antibodies (Ipilimumab) block the inhibitory signals from the T cell to the Cancer cell, leading to T cell activation and subsequent Cancer cell proliferation. Anti-PD-L1 antibodies (Atezolizumab, Avelumab, Durvalumab) block the interaction between PD-L1 on the Cancer cell and PD-1 on the T cell, also leading to T cell activation and Cancer cell proliferation.

Adverse Effects of Immune Checkpoint Inhibitors

Common Adverse Effects

- ✦ Diarrhea
- ✦ Fatigue
- ✦ Cough
- ✦ Nausea
- ✦ Skin rash
- ✦ Poor appetite
- ✦ Constipation
- ✦ Muscle and joint pain



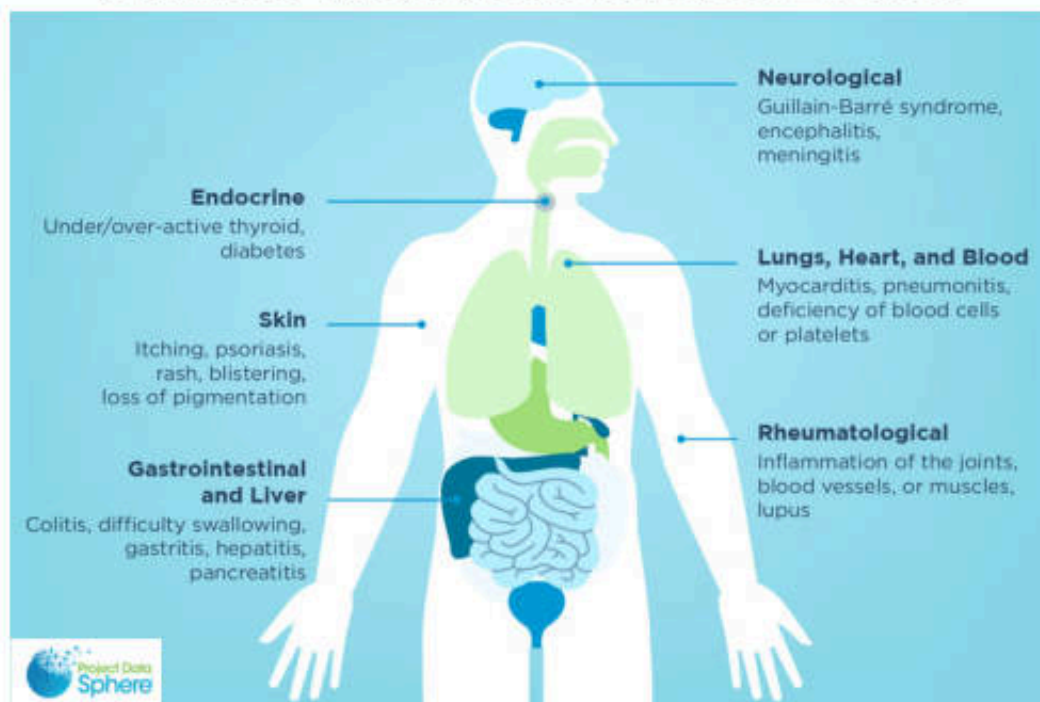
Serious Adverse Effects

- ✦ **Infusion reaction**
: fever, chills, flushing of the face, rash, itchy skin, feeling dizzy, wheezing, and trouble breathing
- ✦ **Immune-related adverse events (irAEs)**
: the immune system attack other parts of the body, including lungs, intestines, liver, hormone-making glands, and kidneys



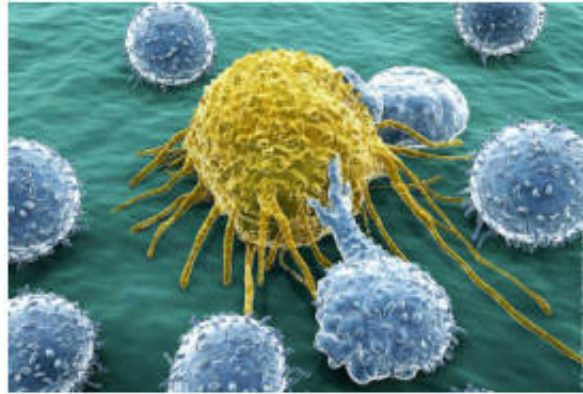
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Immune-Mediated Adverse Events



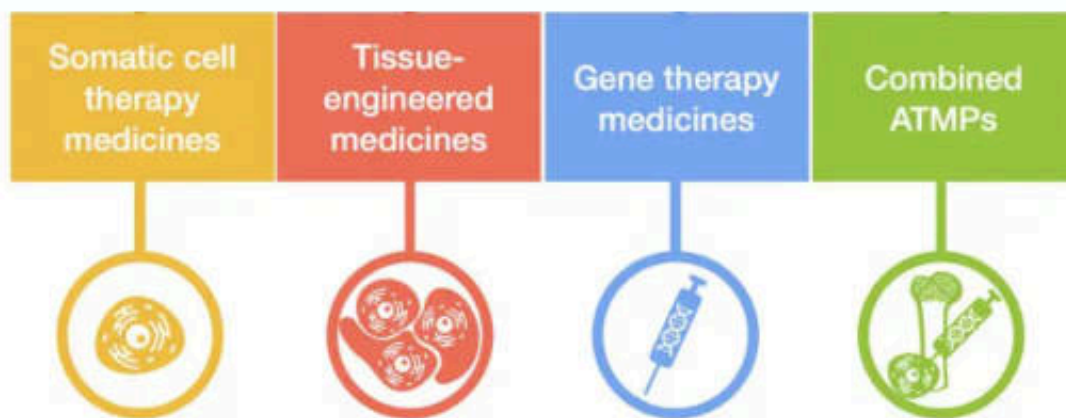
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CAR T-Cell Therapy



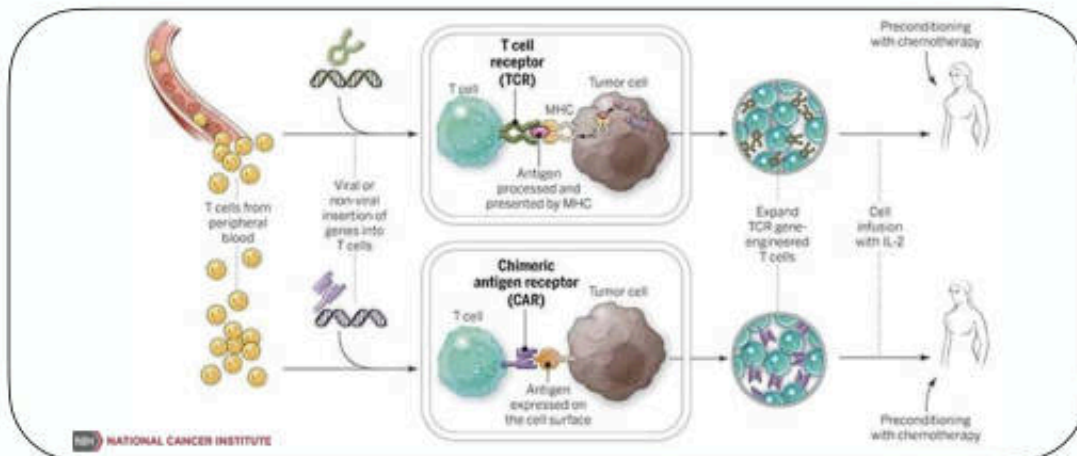
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Advanced Therapy Medicinal Products (ATMPs)



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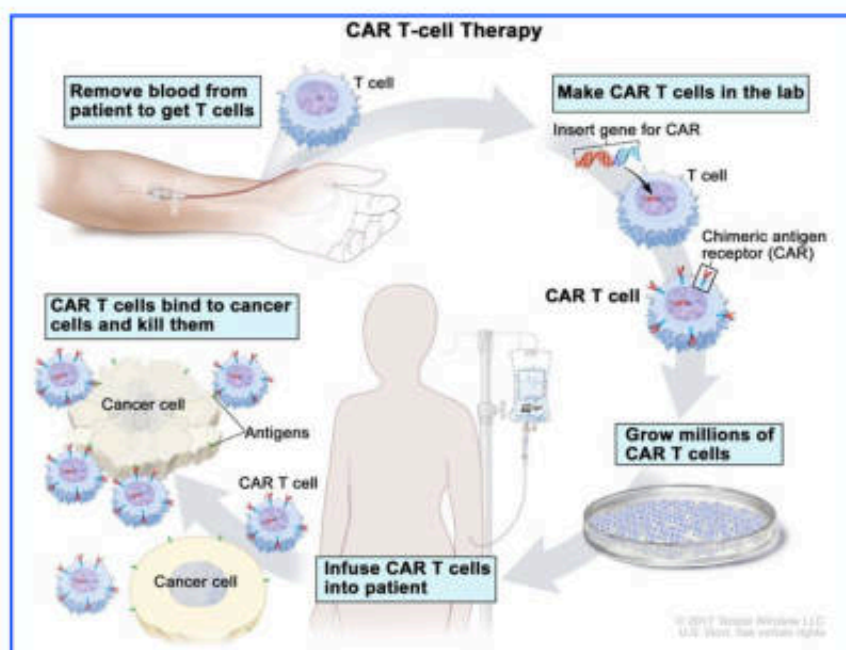
Chimeric Antigen Receptor T-Cell (CAR T-Cell) Therapy



- ♦ A type of treatment in which a patient's T cells are changed in the laboratory so they will attack cancer cells
- ♦ The gene for a special receptor, called a chimeric antigen receptor (CAR), that binds to a certain protein on the patient's cancer cells is added in the laboratory

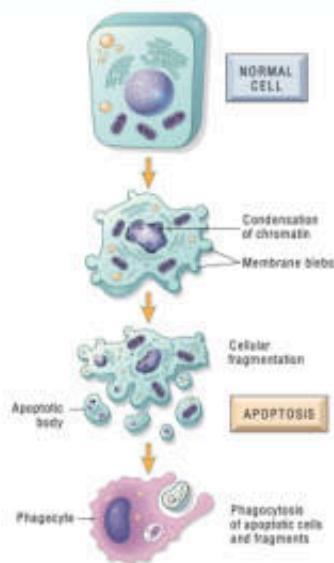
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CAR T-Cell Therapy



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Apoptosis-Inducing Agents

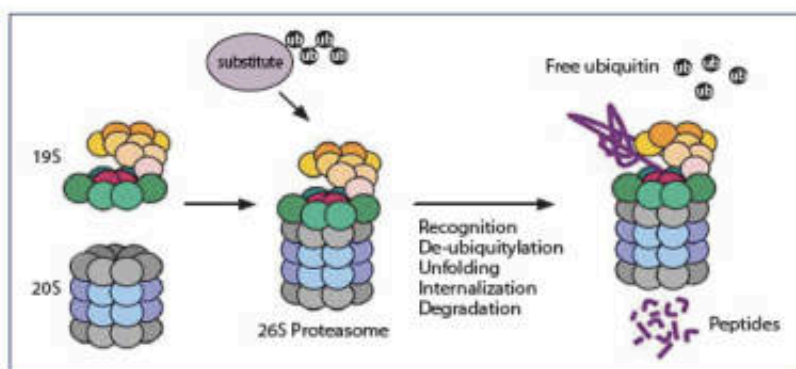


- ♦ The process of programmed cell death
- ♦ It occurs normally during development and aging and as a homeostatic mechanism to maintain cell populations in tissues

Robbins and Cotran pathologic basis of disease, 9th ed.

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Proteasome



- ➡ Plays a key role in maintaining normal cellular homeostasis in eukaryotic cells
- ➡ It regulates various intracellular proteins, including those controlling cell-cycle progression, apoptosis, DNA repair, and the stress response....*such as NF- κ B*
- ➡ Cancer cells generally have higher levels of proteasome activity compared with normal well-differentiated cell

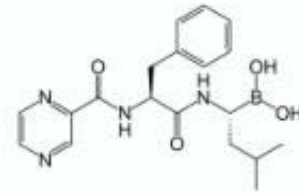
82

Oncology (Williston Park). 2011;25 Suppl 2:19-24.

Proteasome Inhibitors

Bortezomib (Velcade®)

- ⇒ binds to the $\beta 5$ subunit of the 20S core of the 26S proteasome and reversibly inhibits its chymotrypsin-like activity
- ⇒ disrupting multiple intracellular signaling cascades
- ⇒ leading to apoptosis



✦ Elimination

: CYP3A4 and CYP2D6

✦ Adverse effects

: Common

- ⇒ Mild maculopapular rash, mucositis, anemia, and fatigue
- ⇒ Each occurring in 30-50% of patients

: Less common

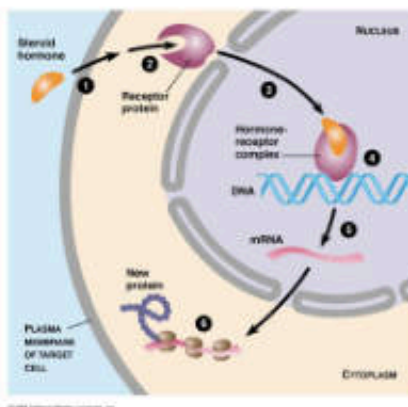
- ⇒ Hyperglycemia, hypertriglyceridemia

Critical targets

- NF- κ B inhibition
- ↑ pro-apoptotic proteins
- Impairment of DNA-damage response
- Cell-cycle arrest
- Inhibition of angiogenesis
- ER stress and unfold protein response

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Hormones and Related Agents

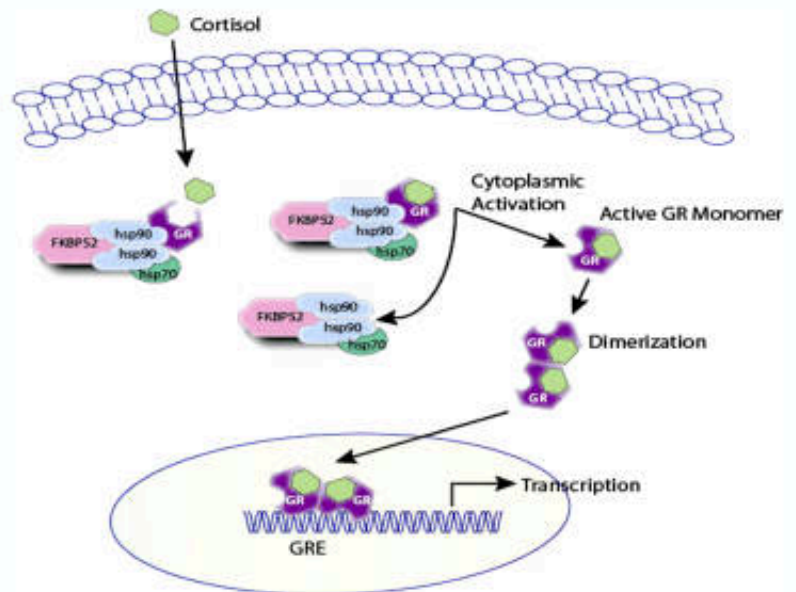


- Glucocorticoids
- Progestins
- Anti-Estrogens (SERMs, SERDs, AIs)
- Anti-Androgens

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Glucocorticoids

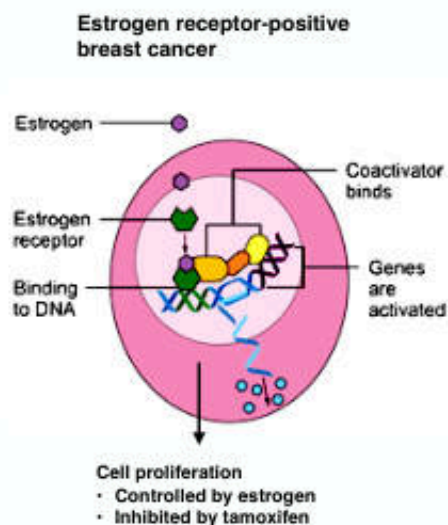
- ❖ Lympholytic effects and their ability to suppress mitosis in lymphocytes
- ❖ Used as cytotoxic agents in the treatment of
 - ⇒ *Acute leukemia in children*
 - ⇒ *Malignant lymphoma in children and adults*



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<http://jmlund.org/blog?p=168>

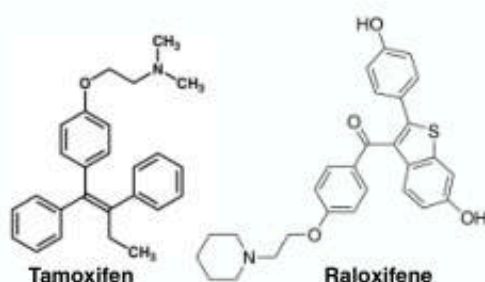
Anti-Estrogens



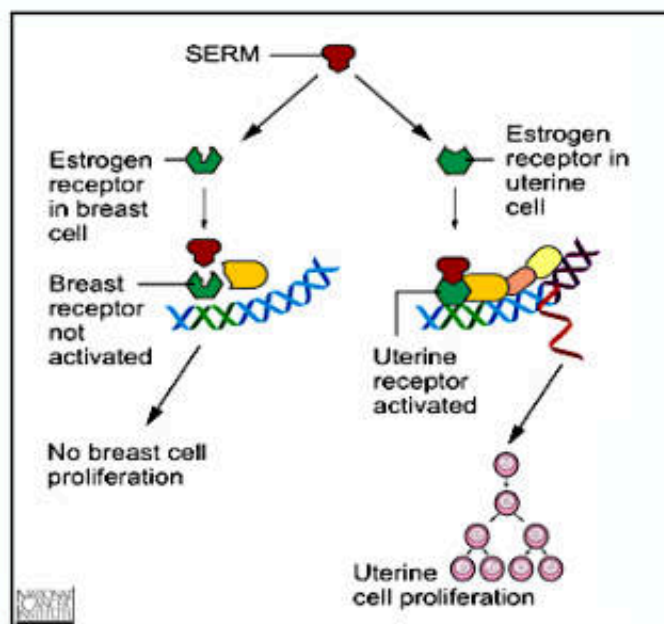
- ❖ Selective estrogen-receptor modulators (SERMs)
- ❖ Selective estrogen-receptor downregulators (SERDs)
- ❖ Aromatase inhibitors (AIs)

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Selective Estrogen-Receptor Modulators (SERMs)



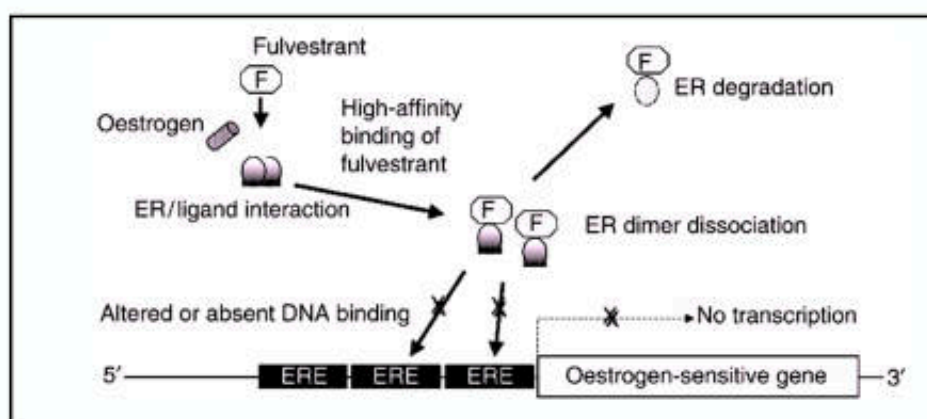
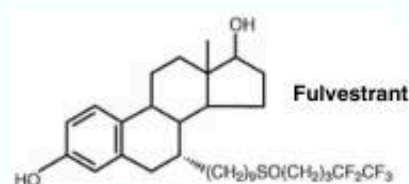
- ♦ Bind to estrogen receptor (ER) and exert either estrogenic or anti-estrogenic effects, depending on the specific organ



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Selective Estrogen-Receptor Downregulators (SERDs)

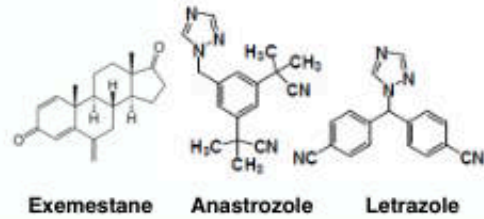
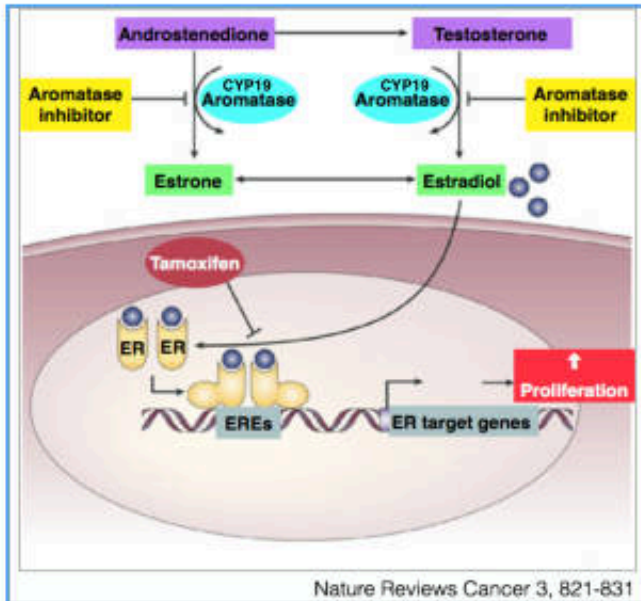
Also termed "pure anti-estrogen"



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British Journal of Cancer (2004) 90, S2-S6

Aromatase Inhibitors (AIs)



Type I: Steroidal AIs

- ⇒ Exemestane
- ⇒ Irreversibly inactivates the enzyme

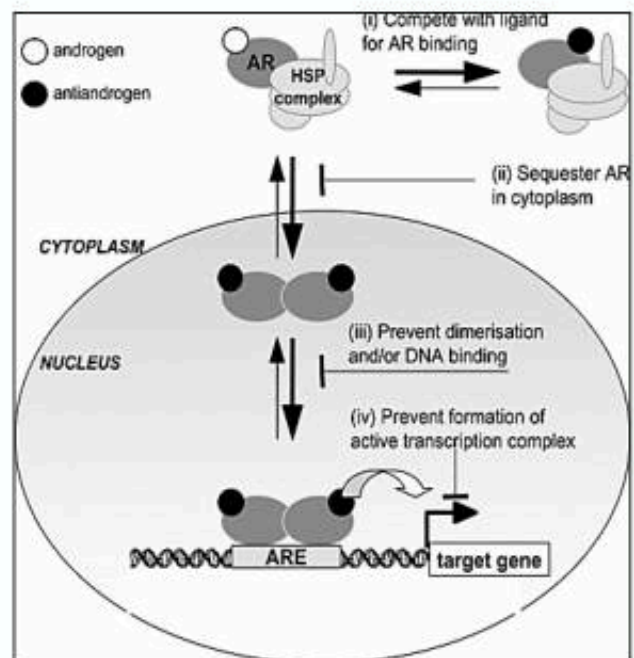
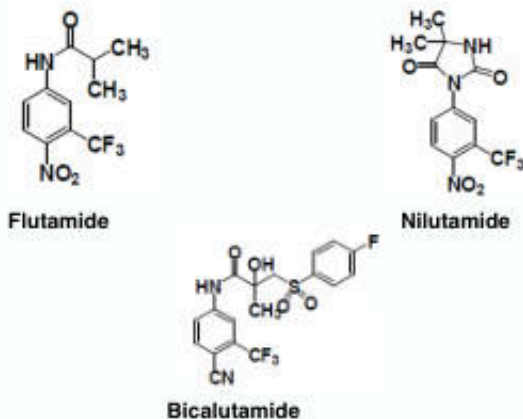
Type II: Nonsteroidal AIs

- ⇒ Anastrozole, letrozole
- ⇒ Reversibly bind to the enzyme
- ⇒ Adjuvant treatment of postmenopausal women with hormone-positive breast cancer, either as initial therapy or sequenced after tamoxifen

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Anti-Androgens

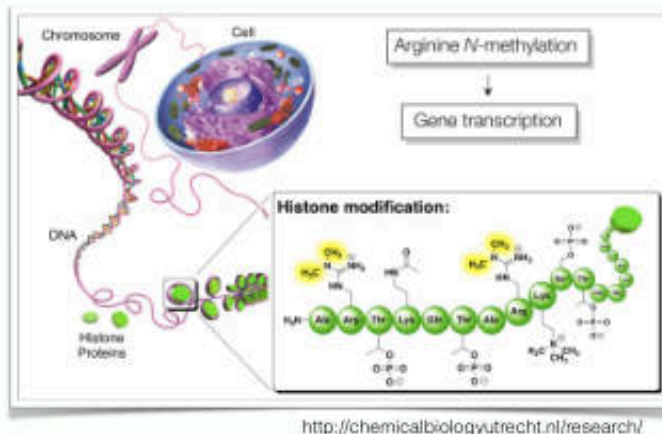
- ✦ Anti-androgen monotherapy is not indicated as first-line treatment for patient with advanced prostate cancer



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Gene Expression Modulators

(Differentiating Agents)

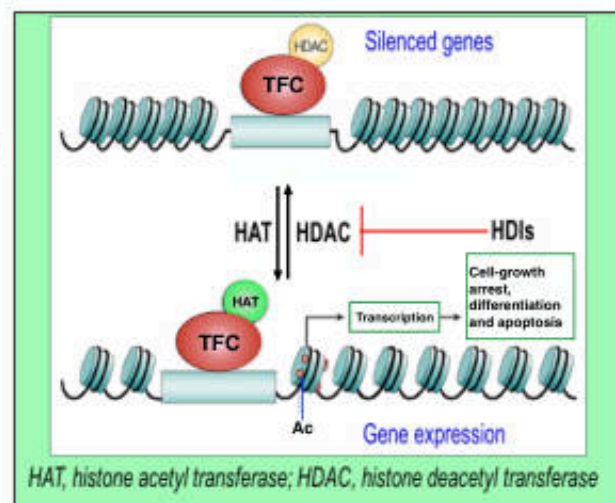


- Histone Deacetylase Inhibitors
- Retinoids
- Arsenic trioxide

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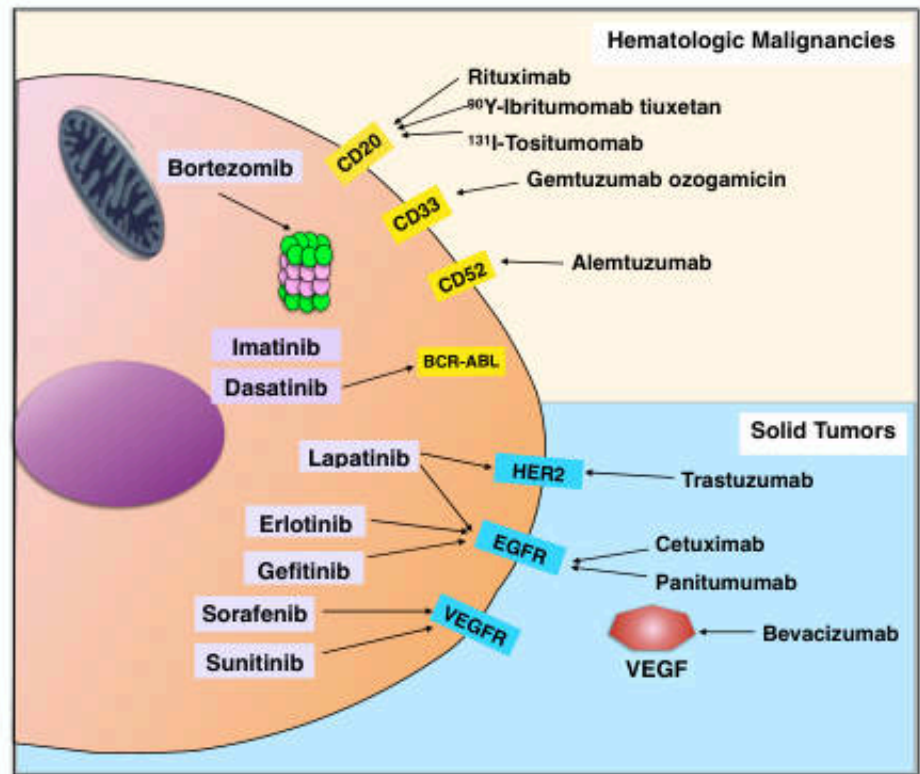
Histone Deacetylase Inhibitors (HDIs)

- ✦ Interfere with the function of histone deacetylase
- ✦ **Vorinostat** and romidepsin is approved to treat cutaneous T-cell lymphoma (CTCL) that has persisted, progressed, or recurred during or after treatment with other medicines



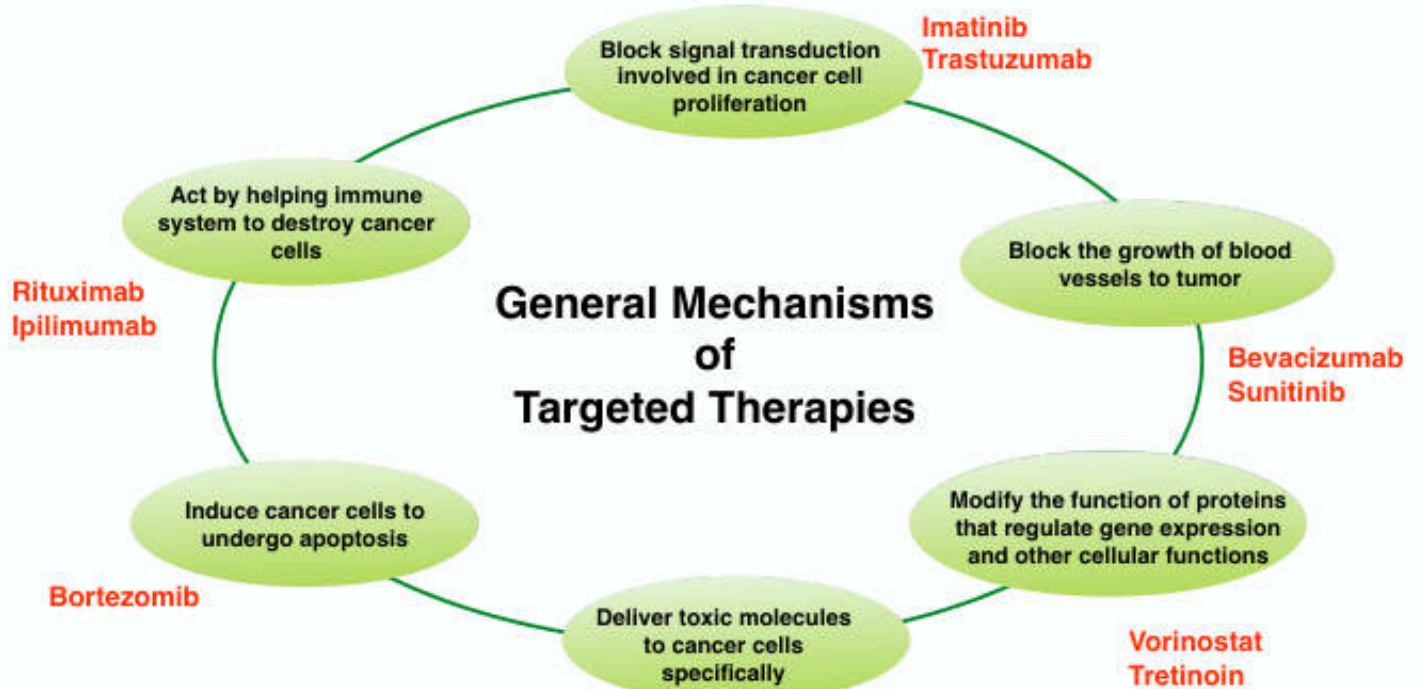
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Targeted Therapy



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CD, Cluster of differentiation



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